

CHECK POINT CERTIFIED SECURITY ADMINISTRATOR

CCSA105 Practical Assignment

Policy Layers

Report detail	Information
Student name	Cossam Kondowe
Student ID	C11/26/CCSA/17044
Course	CCSA105 Policy Layers
Environment	R82 SmartConsole, A-SMS, Security Gateway(s), Internal/DMZ/External test hosts
Policy package	Orion-Layers
Ordered layers	Network; Service-Control
Inline layer	DMZ-Policy

Introduction and Environment

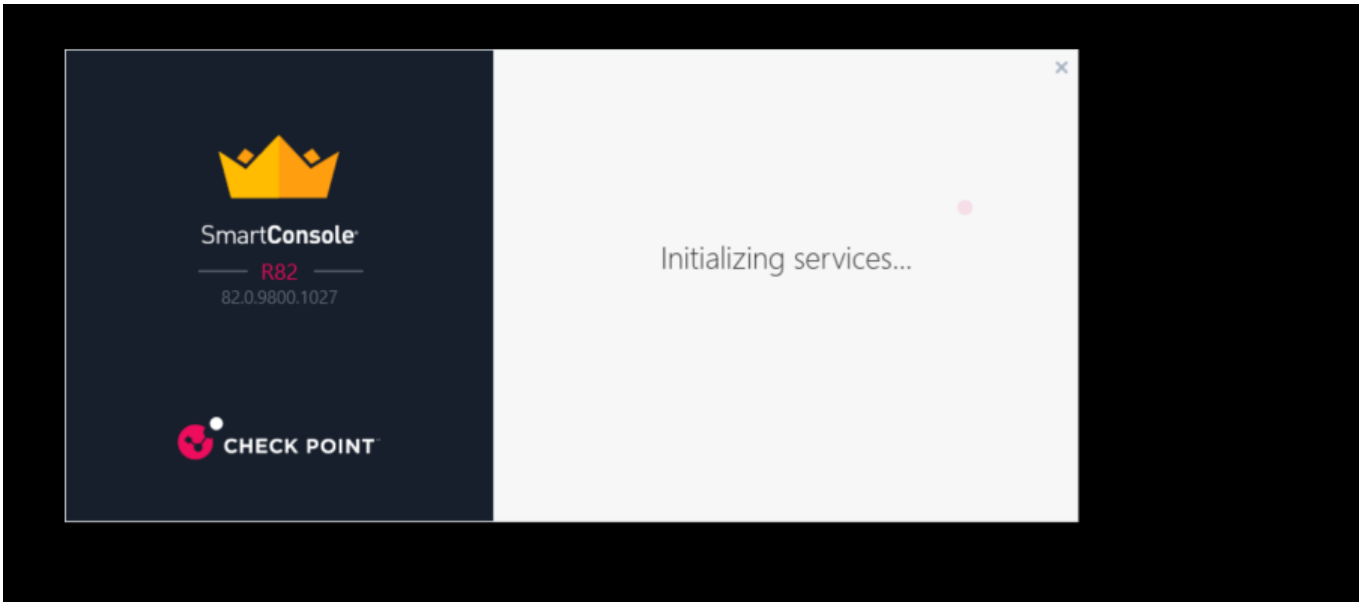
This practical applied Check Point policy-layer concepts in SmartConsole. A policy package named Orion-Layers was created with two Ordered Layers, Network and Service-Control. A shared Inline Layer named DMZ-Policy was then attached to the Internal to DMZ rule. The completed configuration was published, installed on the lab gateway, and verified through traffic tests and log analysis.

Part A: Practical Work

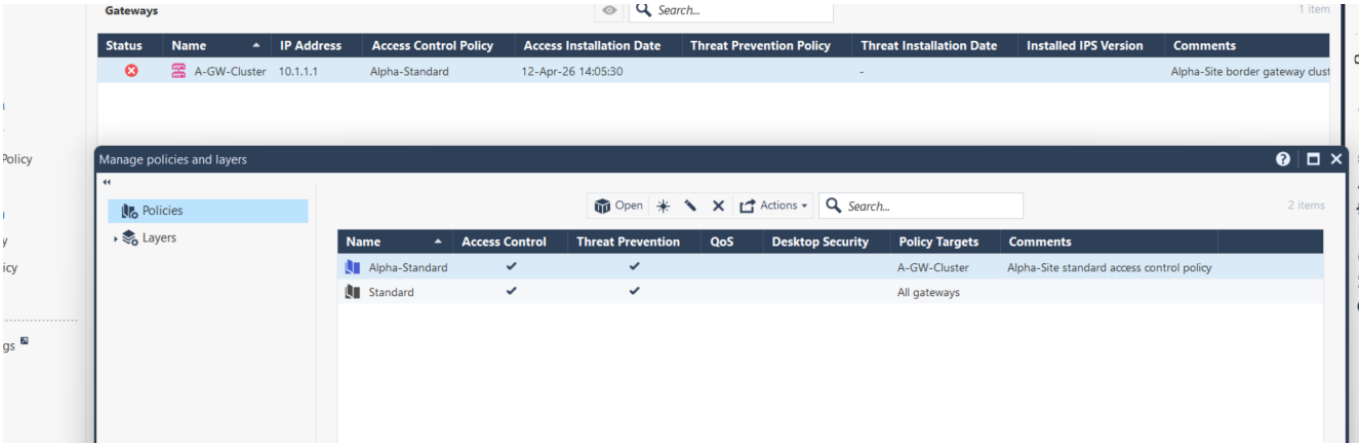
Task 1: Default Policy and Ordered Layer

The Applications menu was opened and Manage policies and layers was selected. The default Access Control policy and its default Ordered Layer were identified. The default layer properties were inspected on the General, Advanced and Permissions pages without changing any settings. The default policy rulebase was then opened and the Cleanup rule action was recorded.

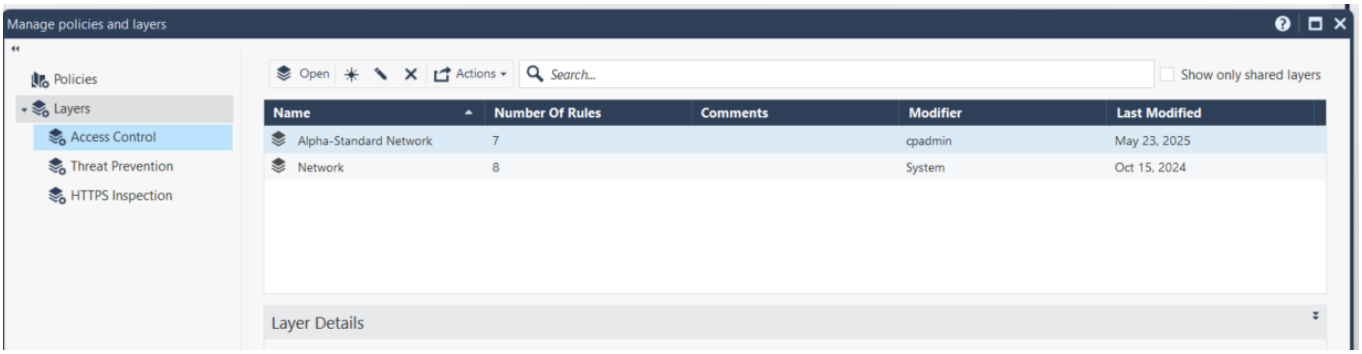
1. Connecting A-SMS using Check Point R82 SmartConsole.

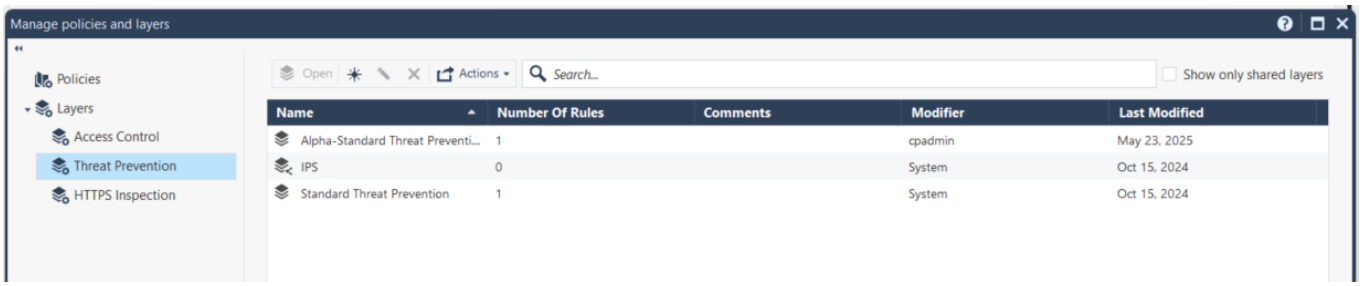


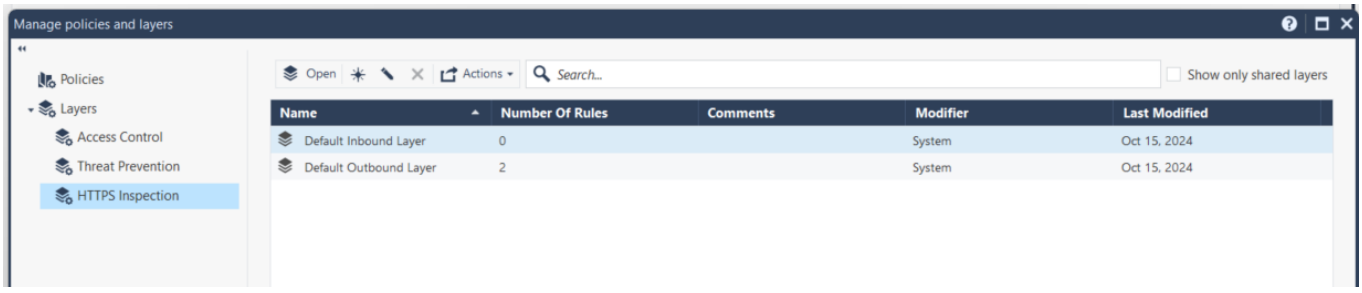
Recording the name of the default Access Control policy.



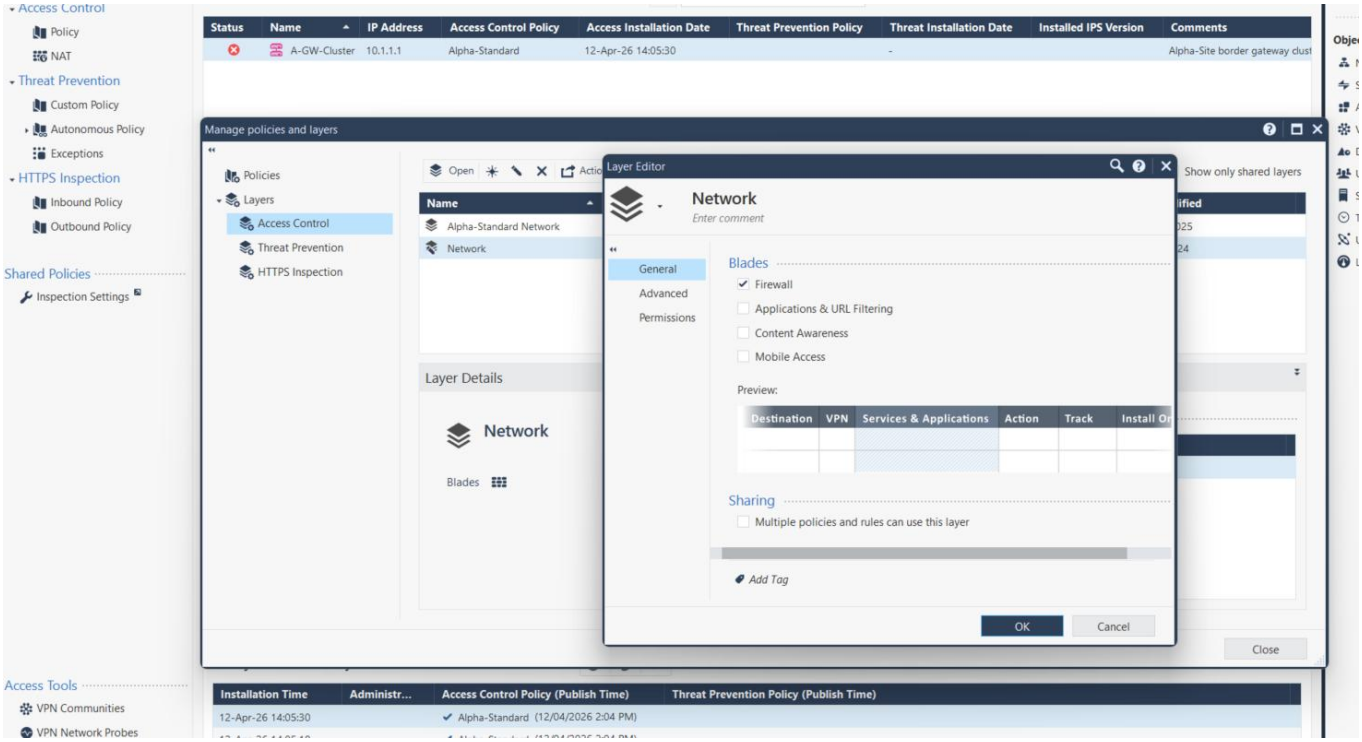
Recording the names of its default Ordered Layer.



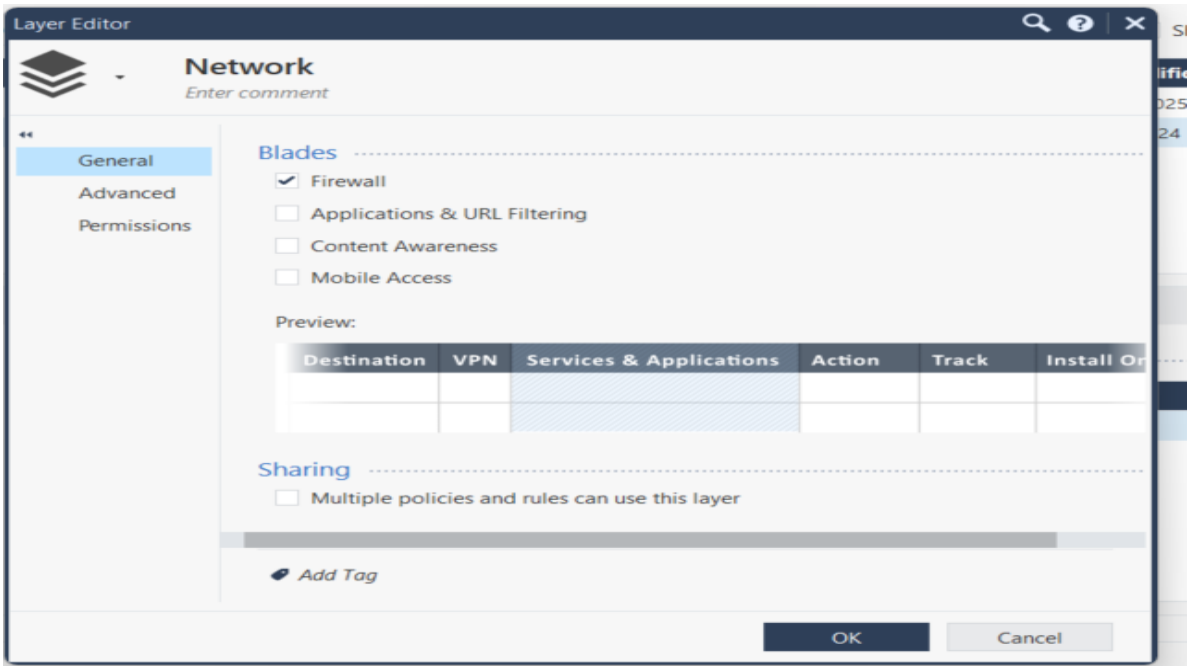




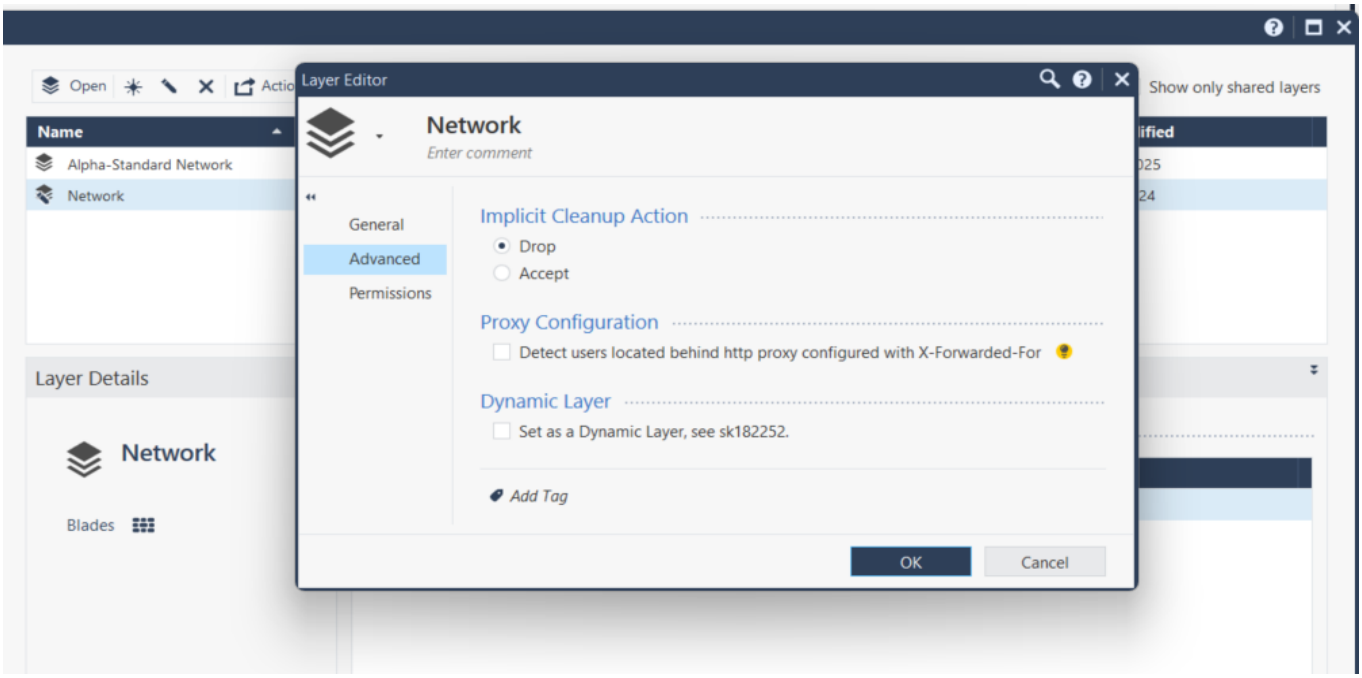
Part 1B: Opening the default Ordered Layer properties



Part 1C: Recording the General settings



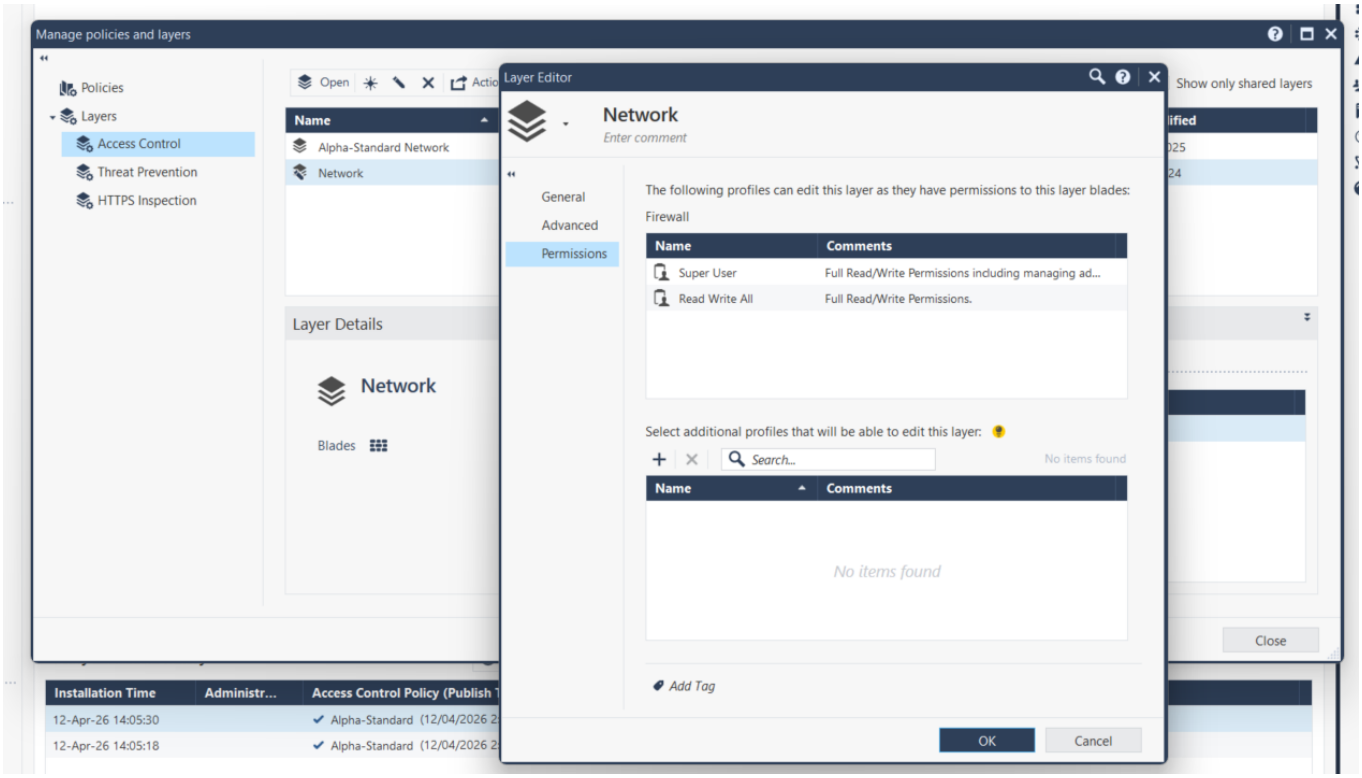
Part 1D: Recording the Advanced settings



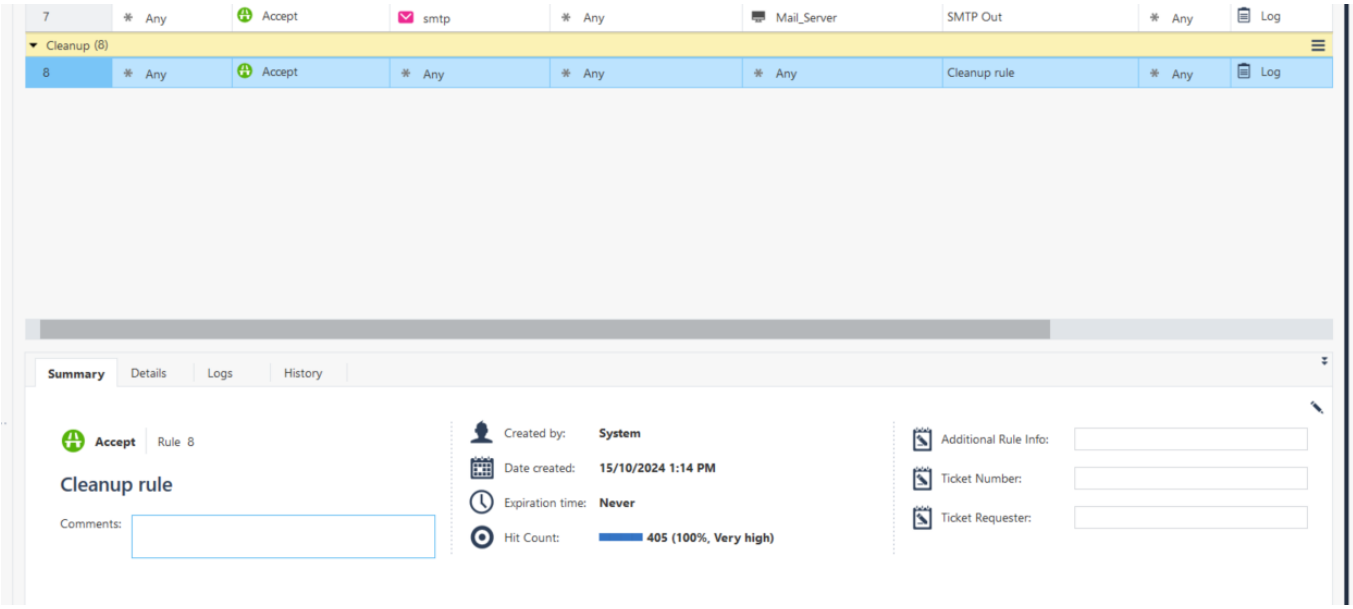
Meaning of Implicit Cleanup Action

The Implicit Cleanup Action applies when traffic reaches the end of the layer without matching an explicit rule. Later tasks require explicit Cleanup rules that use the same action as the corresponding layer's implicit cleanup action.

Part 1E: Recording the Permissions settings



Part 1F: Finding the default Cleanup rule



The default Access Control policy was recorded as Standard and the default Ordered Layer as Network. The layer settings and Cleanup action were documented exactly as displayed in SmartConsole.

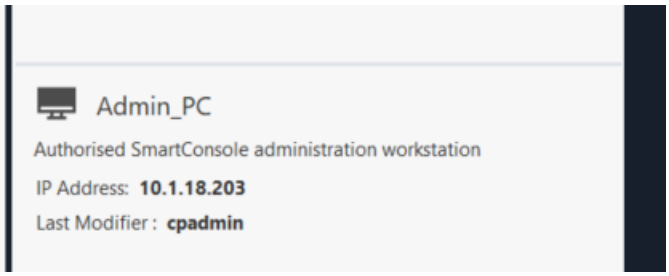
Task 2: Creating Orion-Layers and Building the Network Layer

A. Confirming the required objects exist

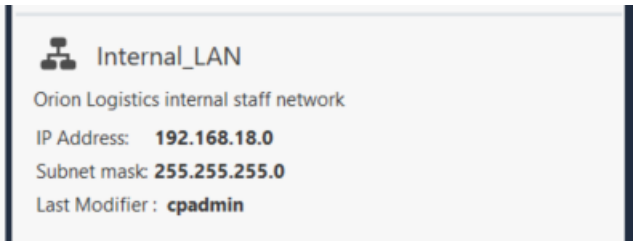
Before creating the package, confirm that these objects are available in the Objects panel:

The objects were created in the previous lab so I just maintained the same

Admin_PC



Internal_LAN

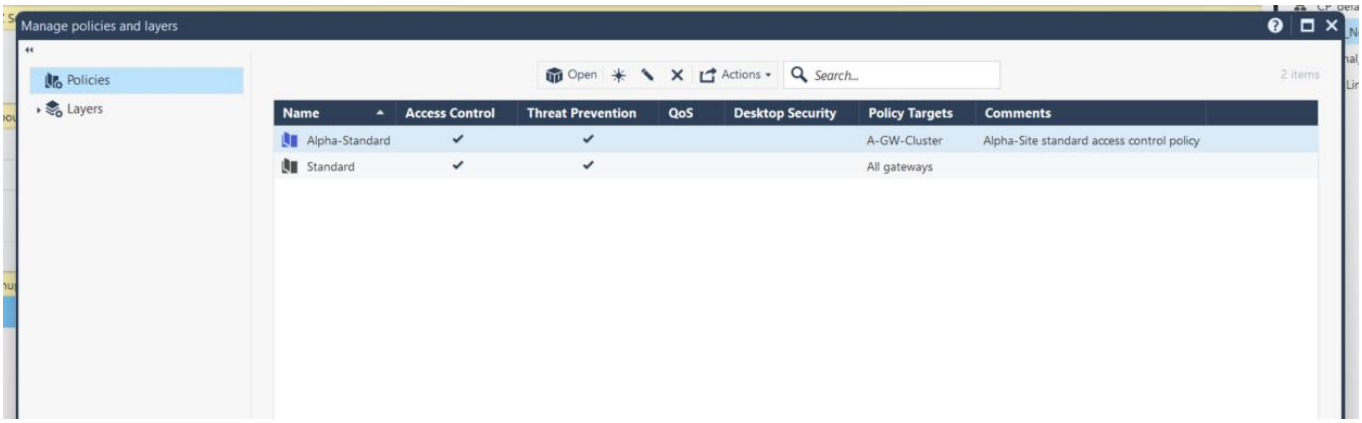


DMZ_NET

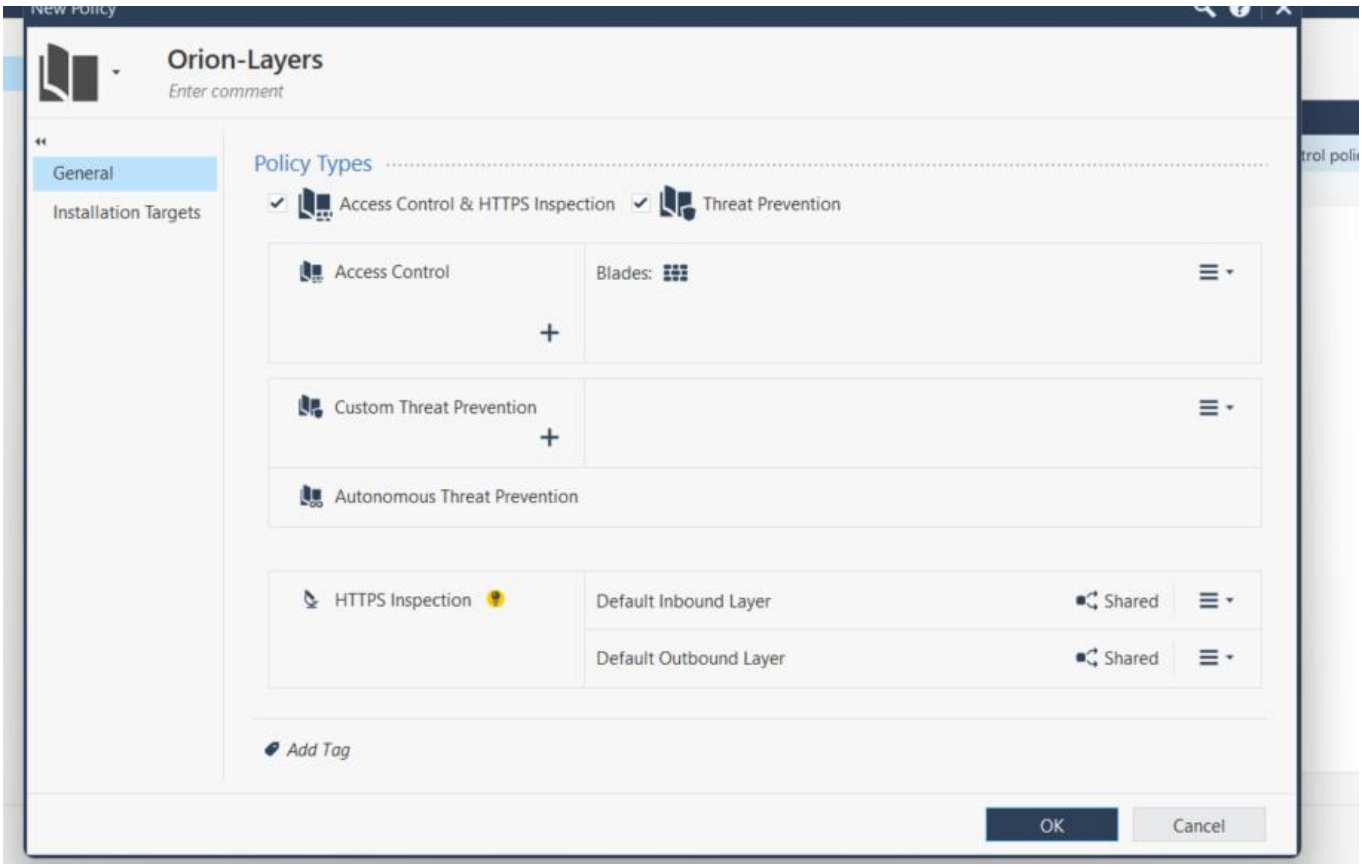


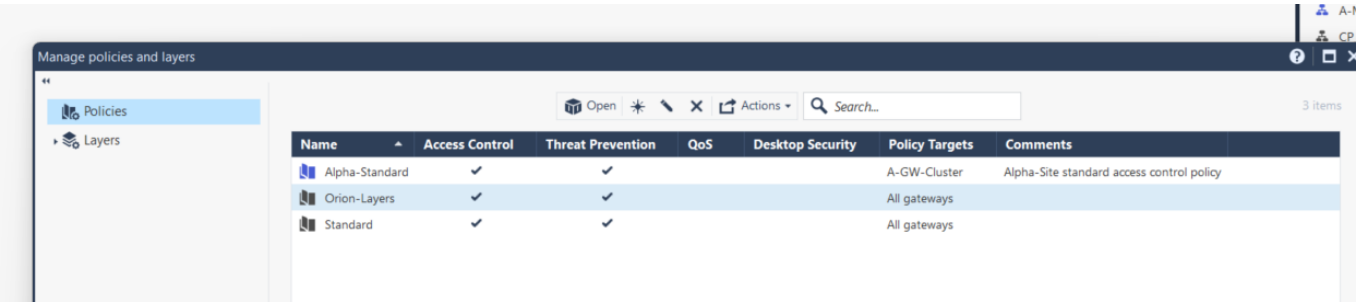
Part 1: Creating the Orion-Layers Policy Package

Step 1: Open Manage Policies and Layers



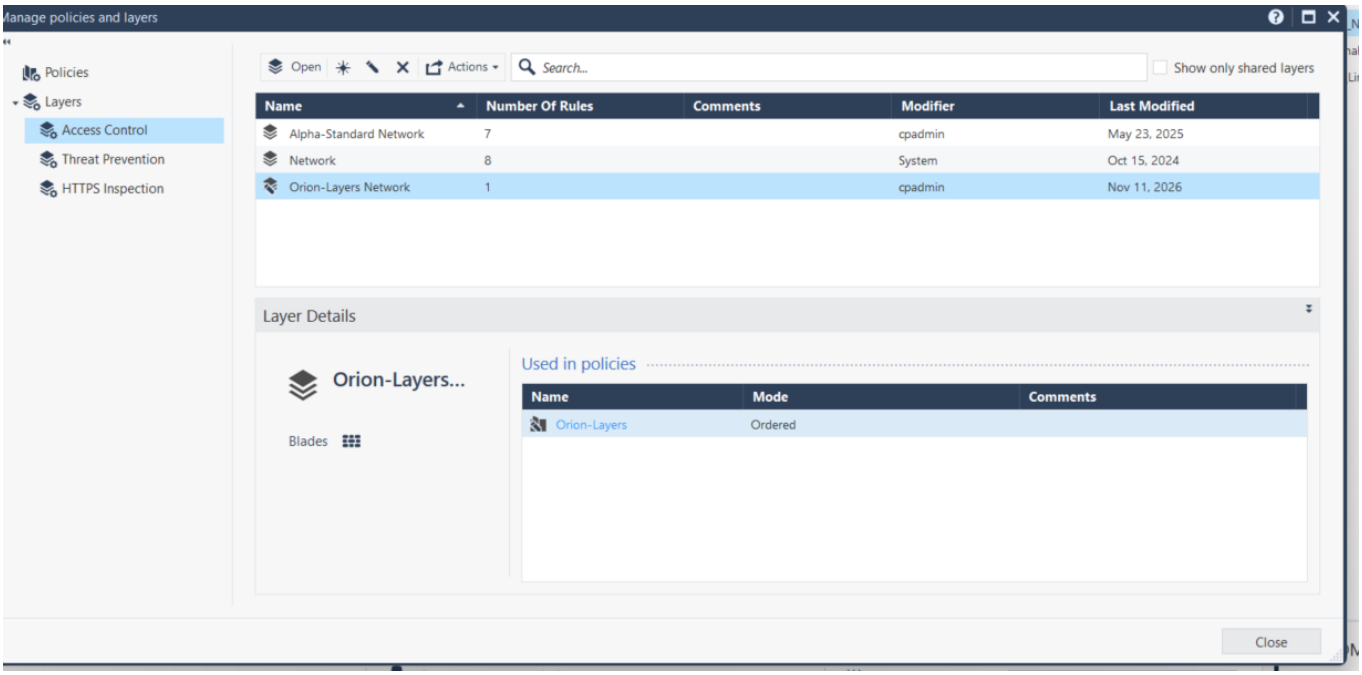
Step 2: The package



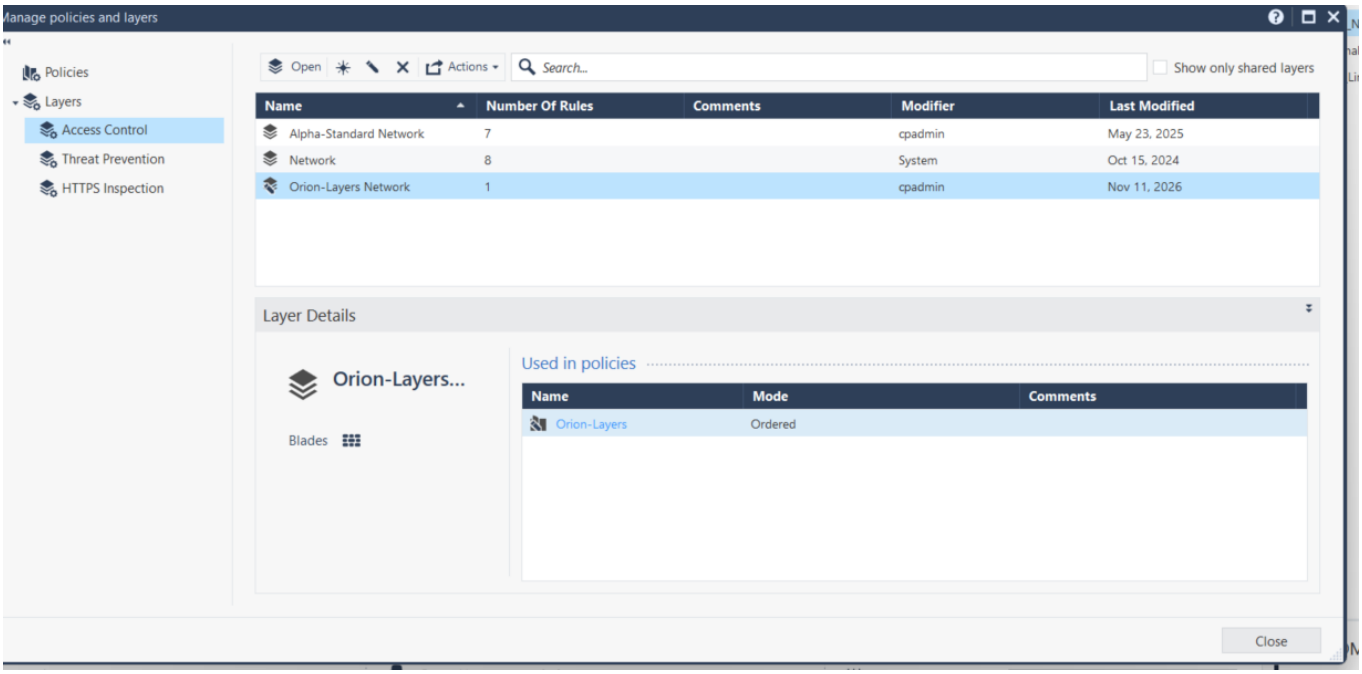


Part 2: Verifying the Ordered Layer

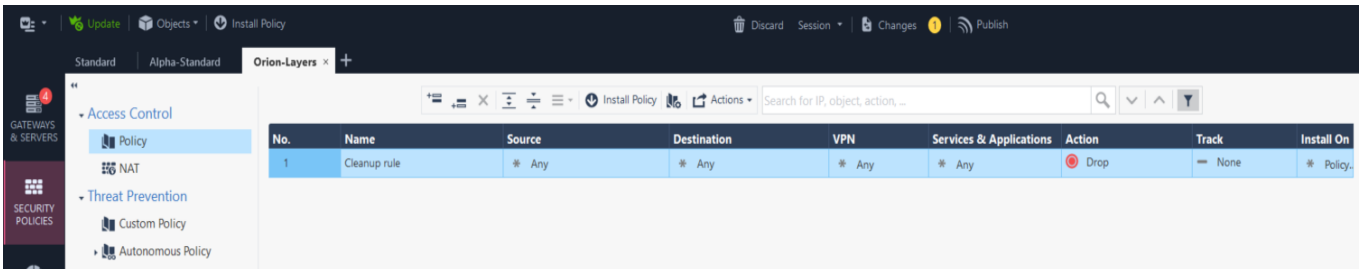
Step 5: Select Orion-Layers



Part 3: Capturing Package Evidence

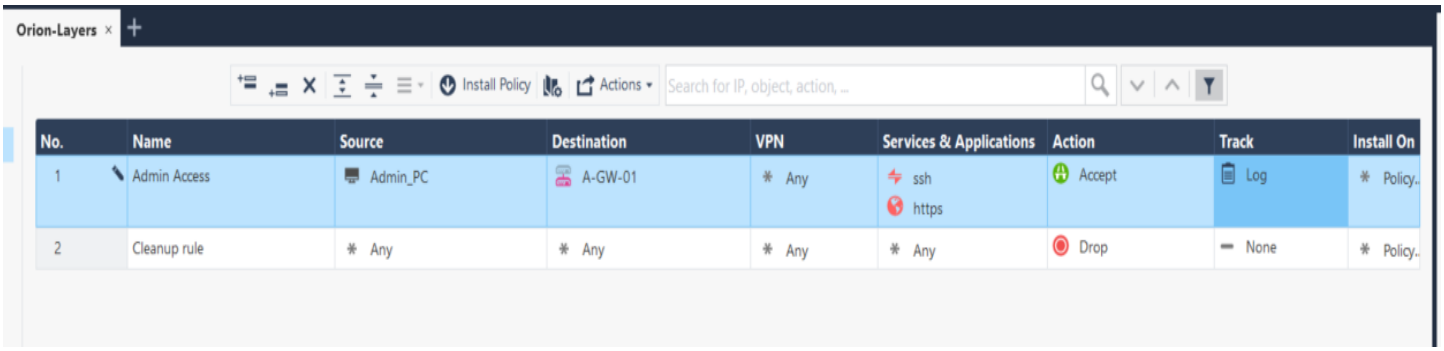


Part 4: Opening the Orion-Layers Rulebase



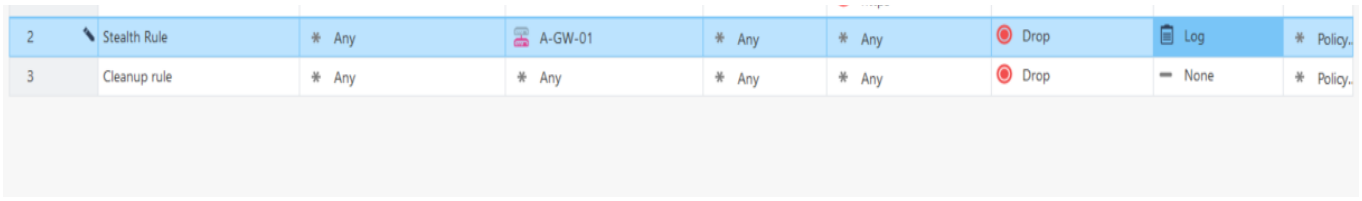
Part 5: Creating the Five Rules

Rule 1: Admin Access



Rule 2: Stealth Rule

Configuring the second rule

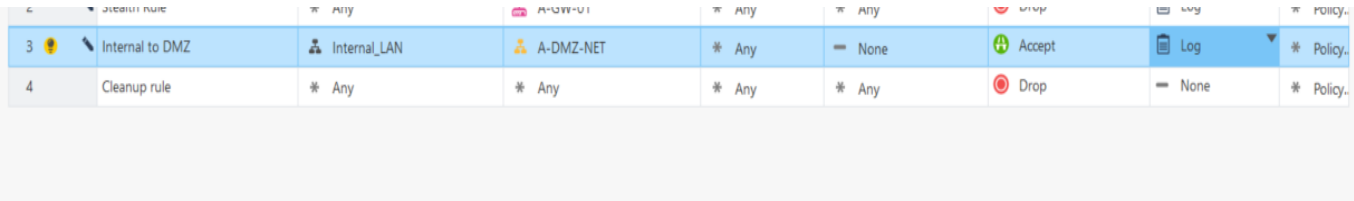


Why the order matters

Admin Access must remain above Stealth Rule.

The first rule permits authorised SSH and HTTPS management traffic. The second rule drops other traffic addressed directly to the gateway.

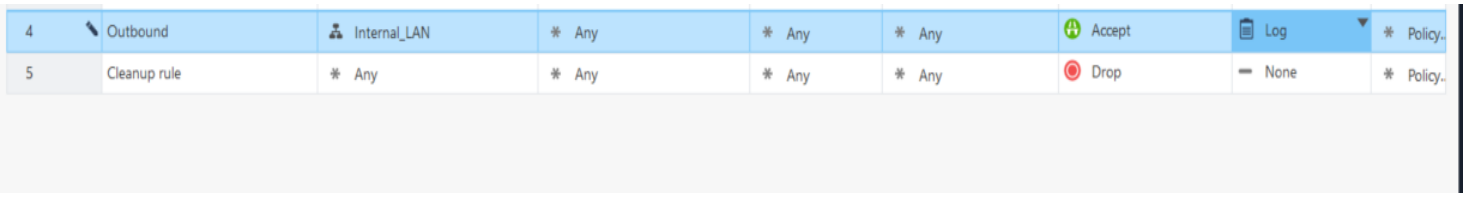
Rule 3: Internal to DMZ



3	Internal to DMZ	Internal_LAN	A-DMZ-NET	Any	None	Accept	Log	Policy..
4	Cleanup rule	Any	Any	Any	Any	Drop	None	Policy..

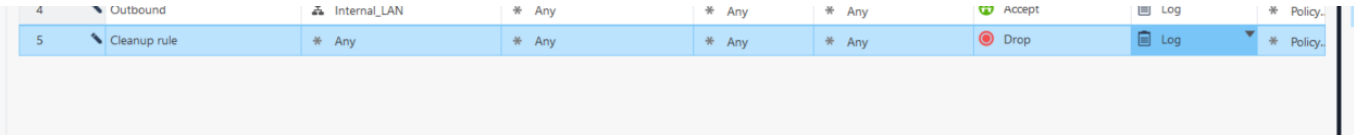
Rule 4: Outbound

Configuring the fourth rule



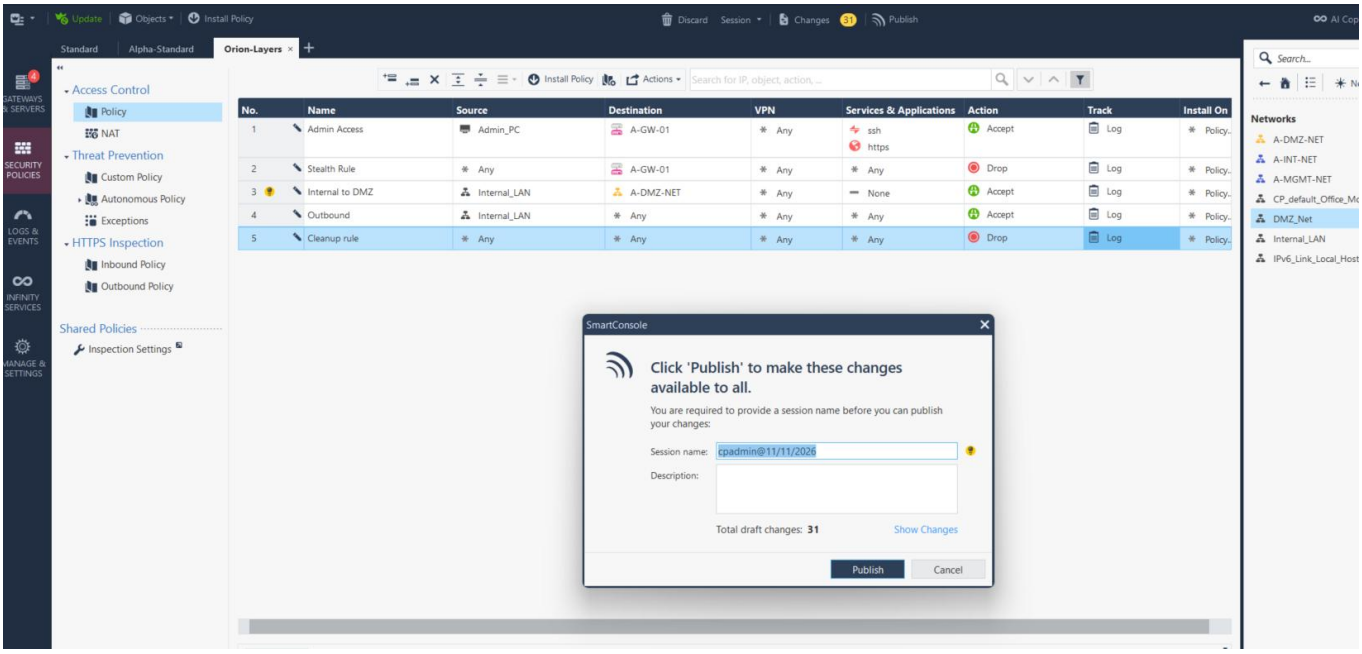
4	Outbound	Internal_LAN	Any	Any	Any	Accept	Log	Policy..
5	Cleanup rule	Any	Any	Any	Any	Drop	None	Policy..

Rule 5: Cleanup Rule

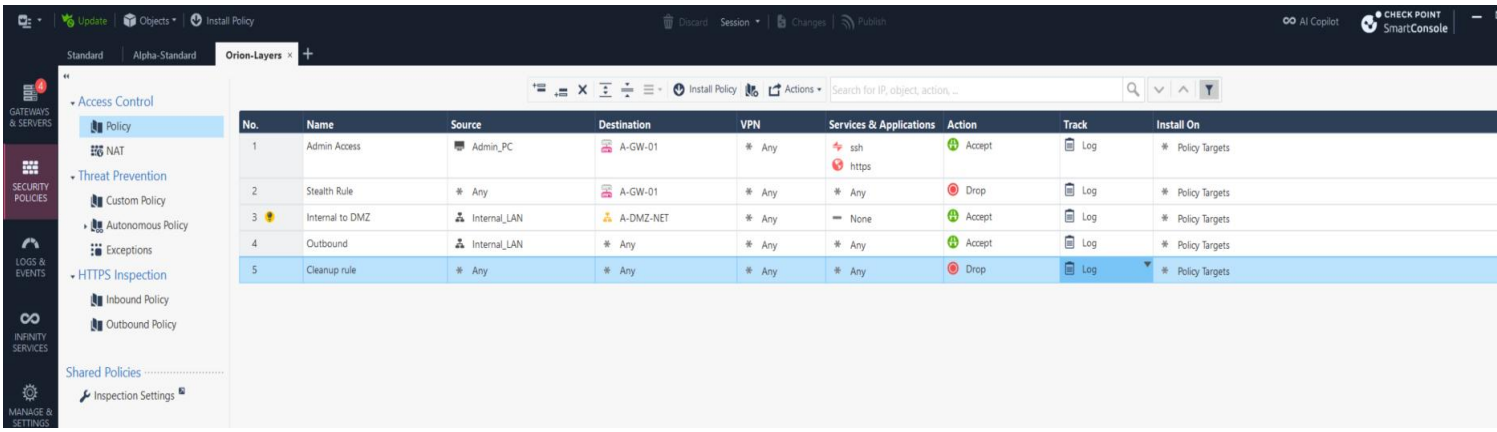


4	Outbound	Internal_LAN	Any	Any	Any	Accept	Log	Policy..
5	Cleanup rule	Any	Any	Any	Any	Drop	Log	Policy..

Part 7: Publishing the Work



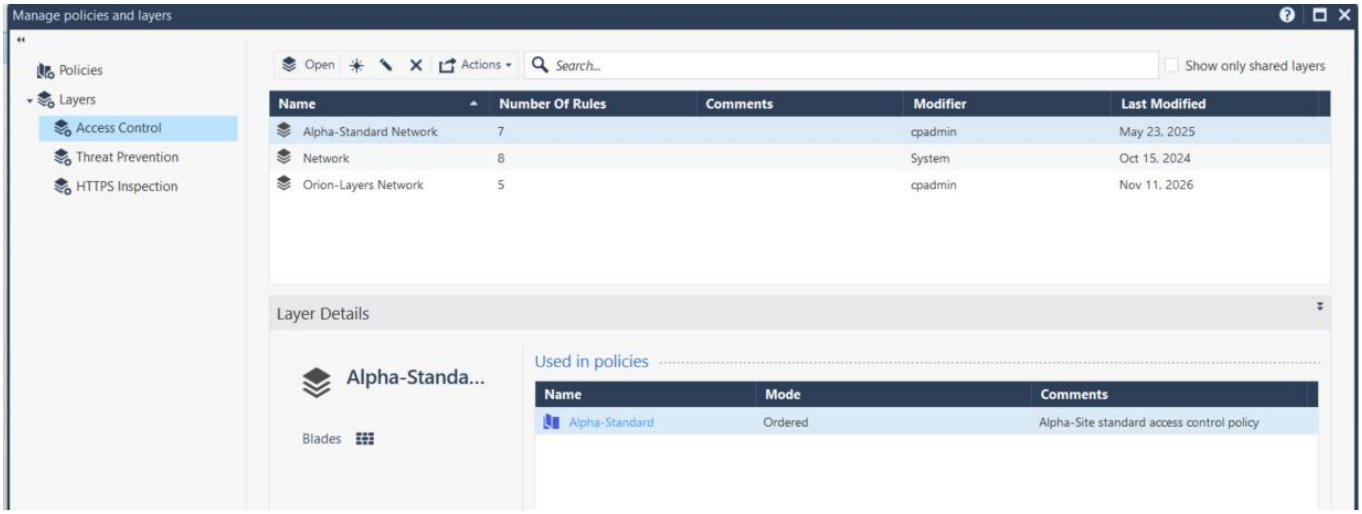
Part 8: Capturing the Full Rulebase Screenshot



Task 3: Adding the Service-Control Ordered Layer

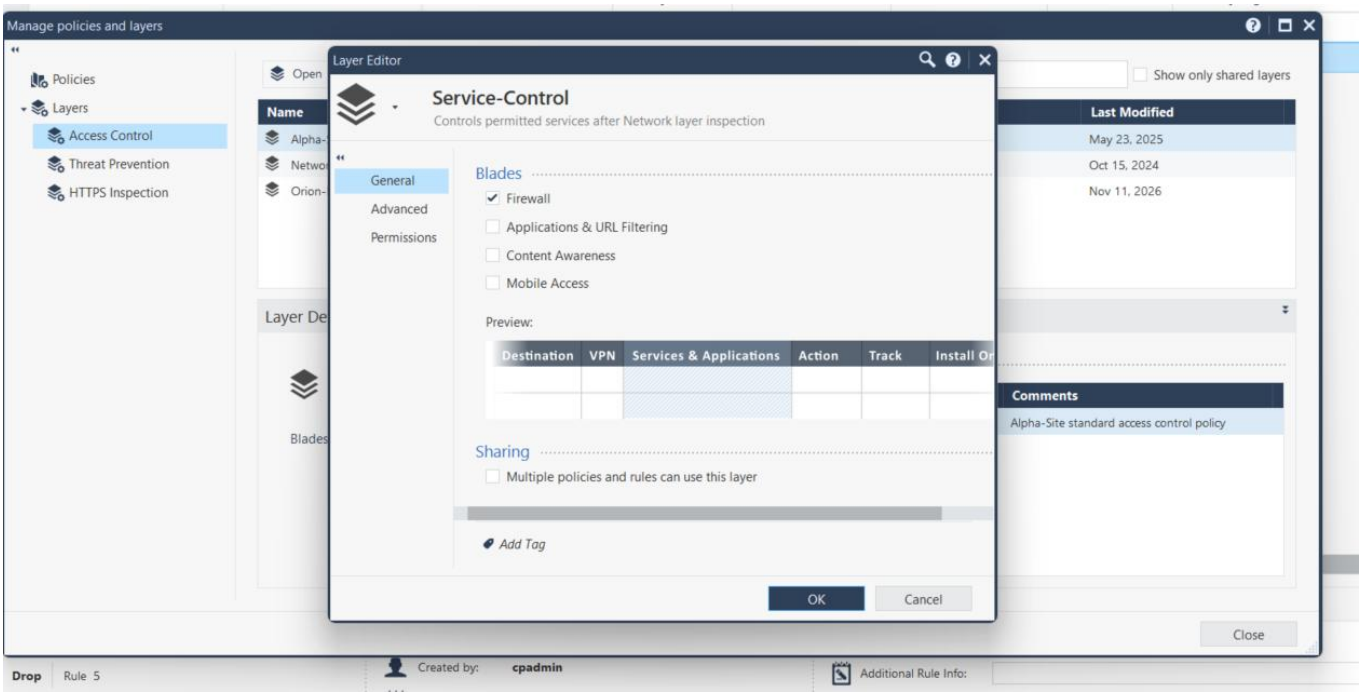
Creating the Service-Control layer

Step 1: Opening Manage Policies and Layers

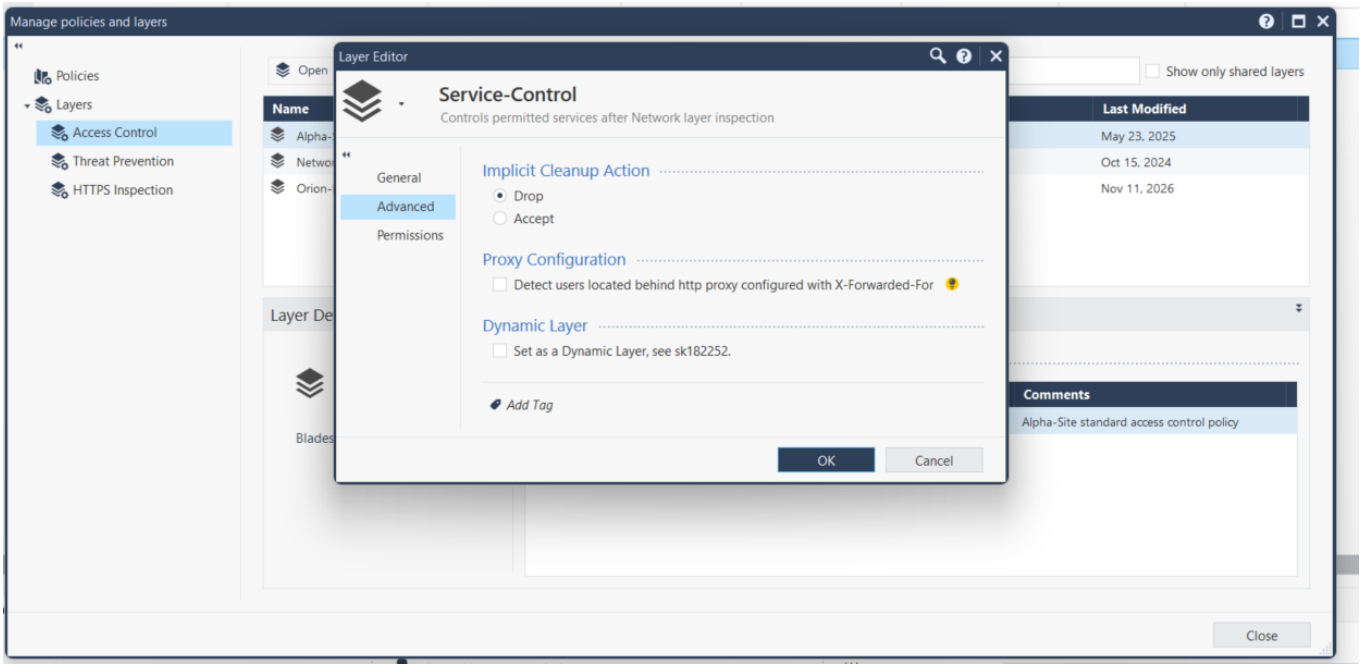


Step 2: Creating a new layer

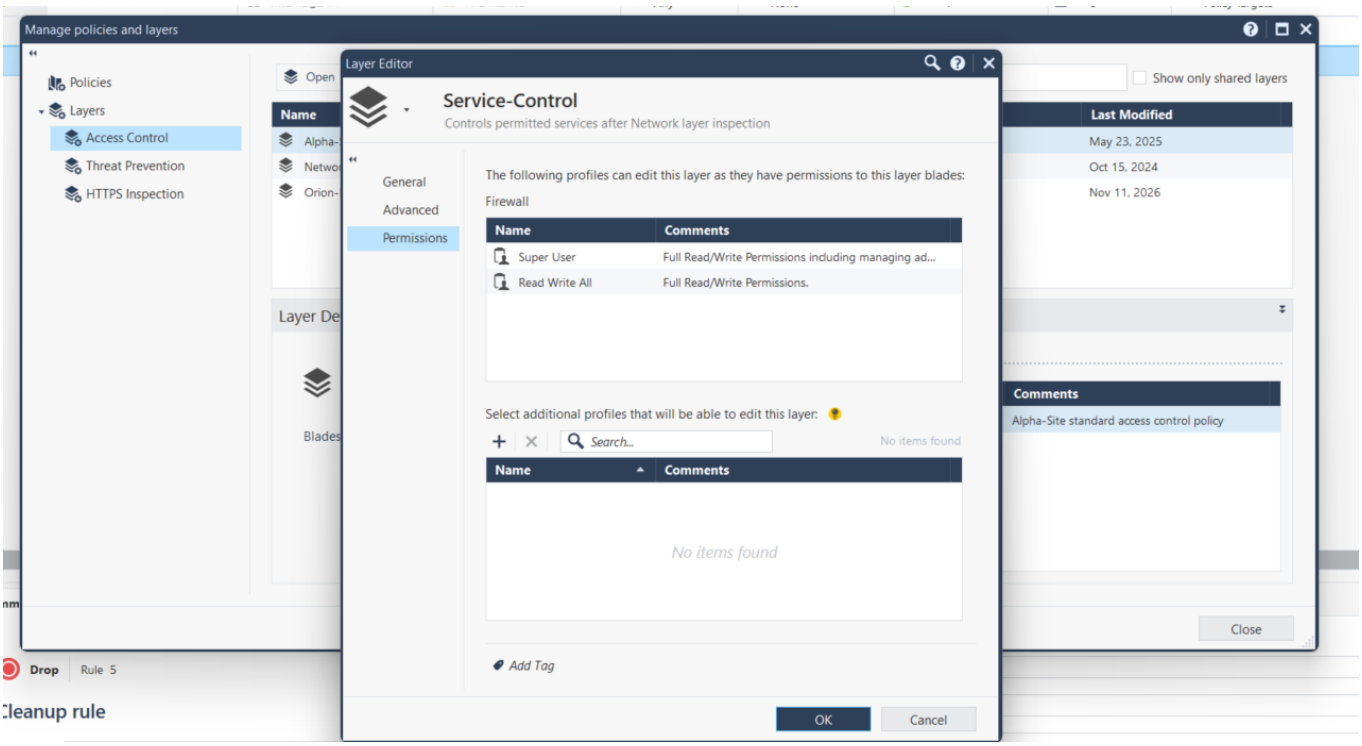
Configuring the General page



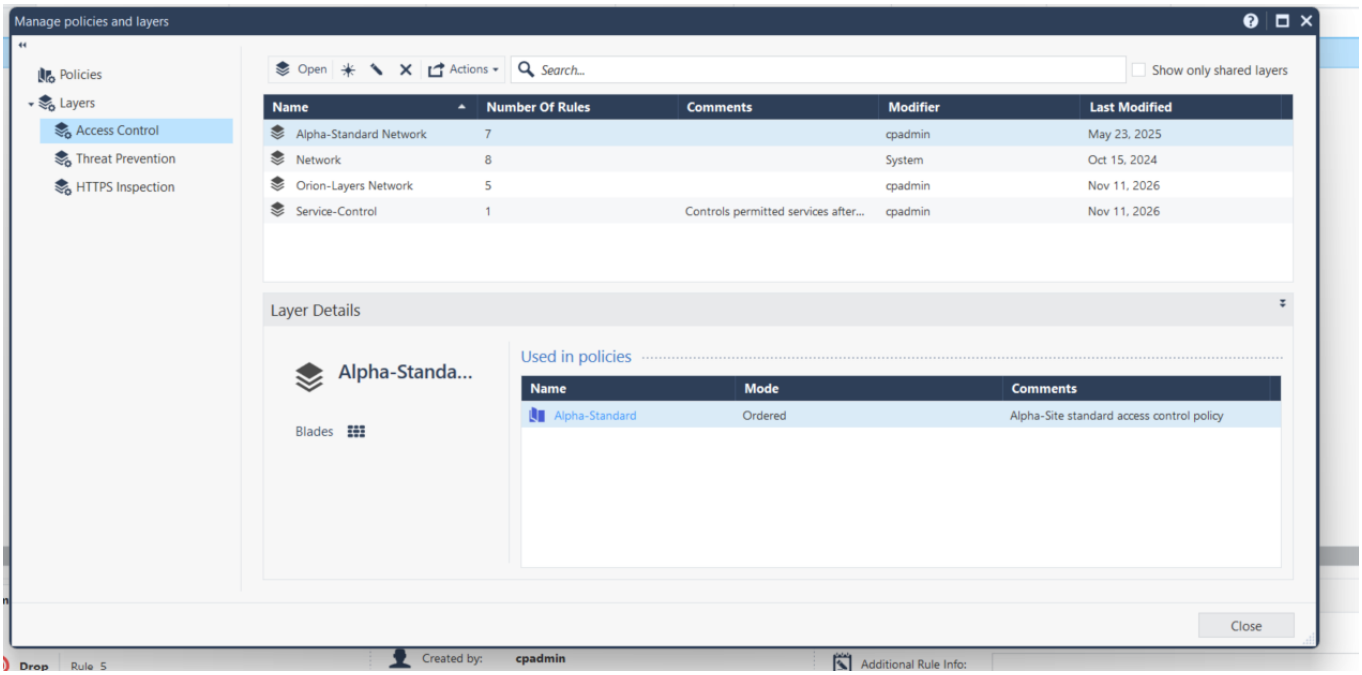
Configuring the Advanced page



Configuring Permissions

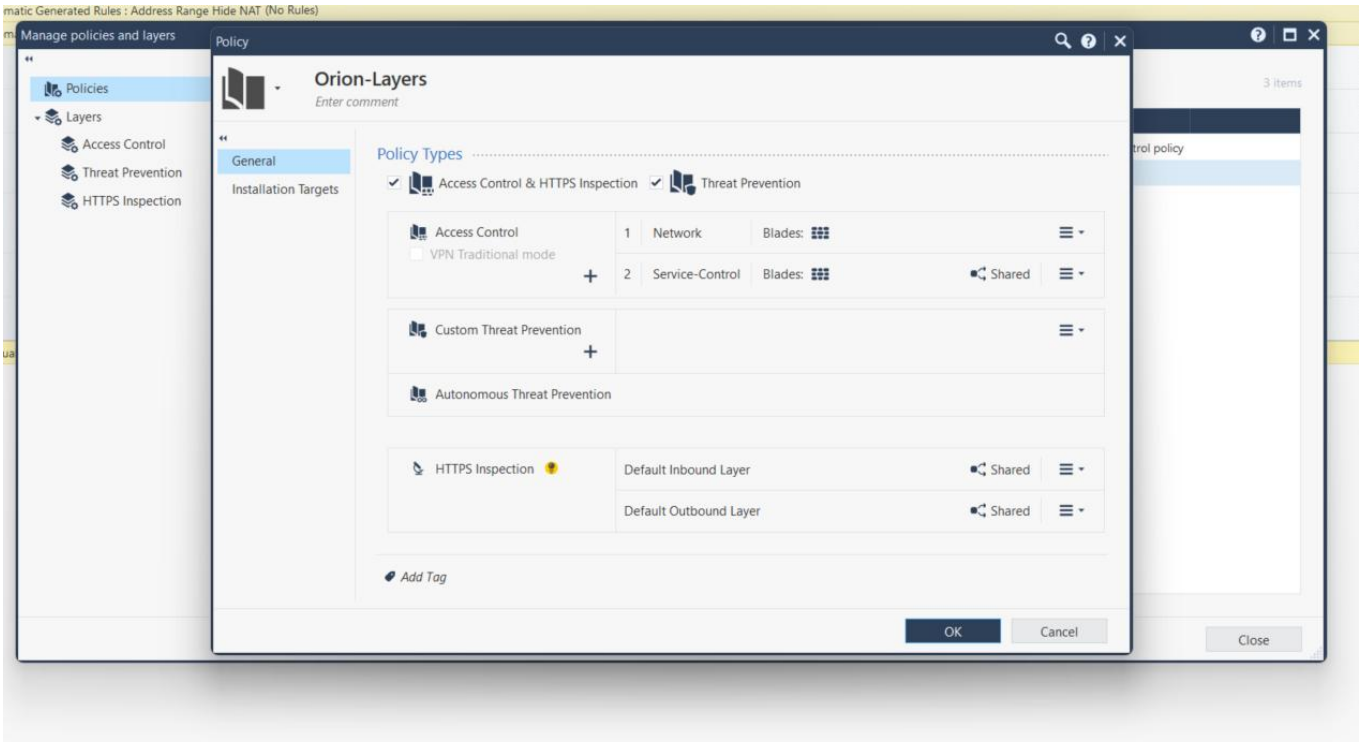


Saving the new layer

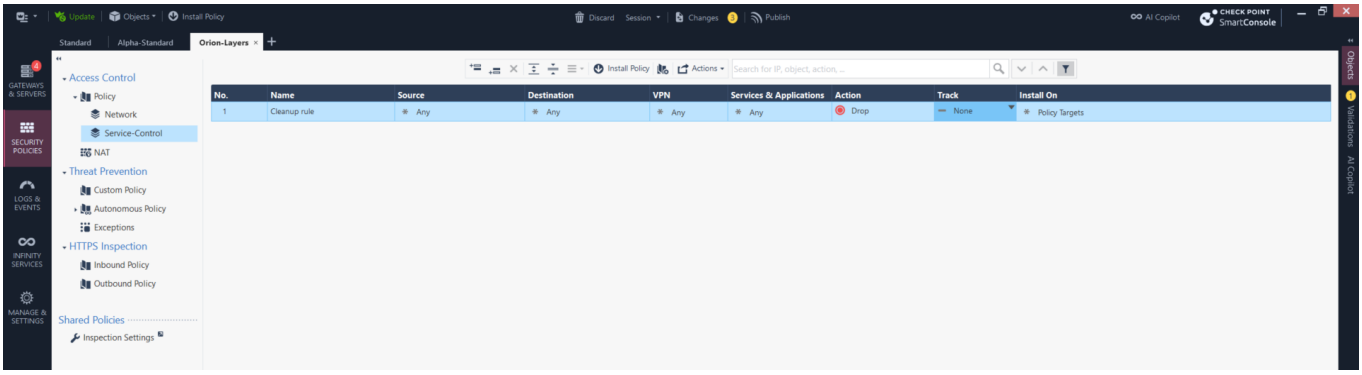


Adding Service-Control to Orion-Layers

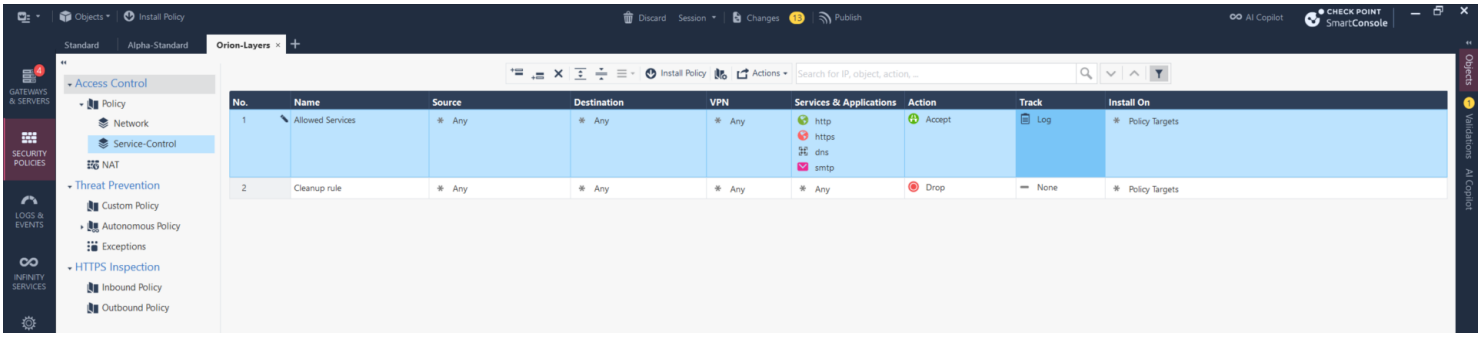
Adding Service-Control



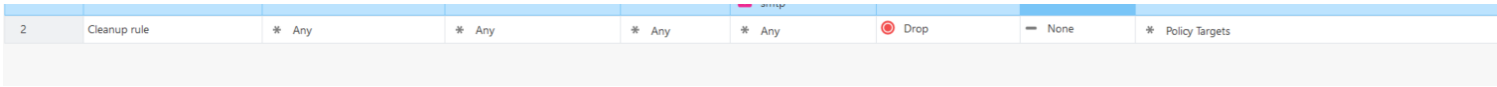
Opening the Service-Control rulebase



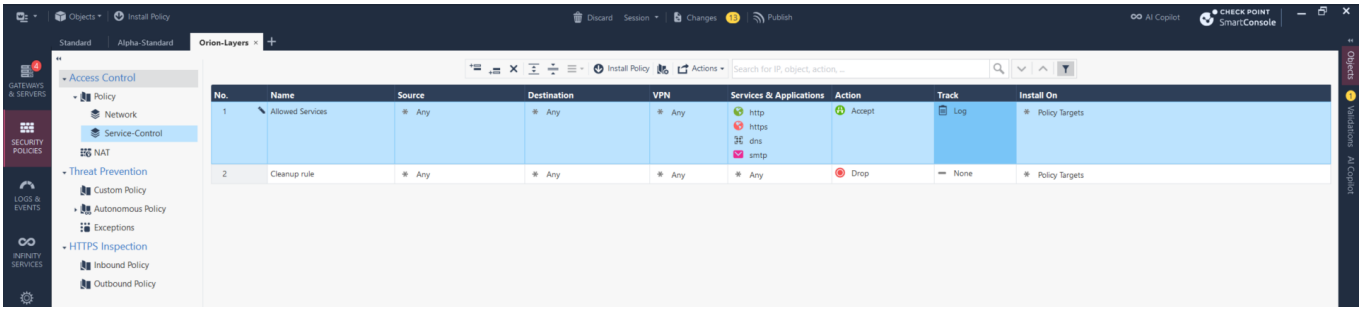
Creating the Service-Control rules



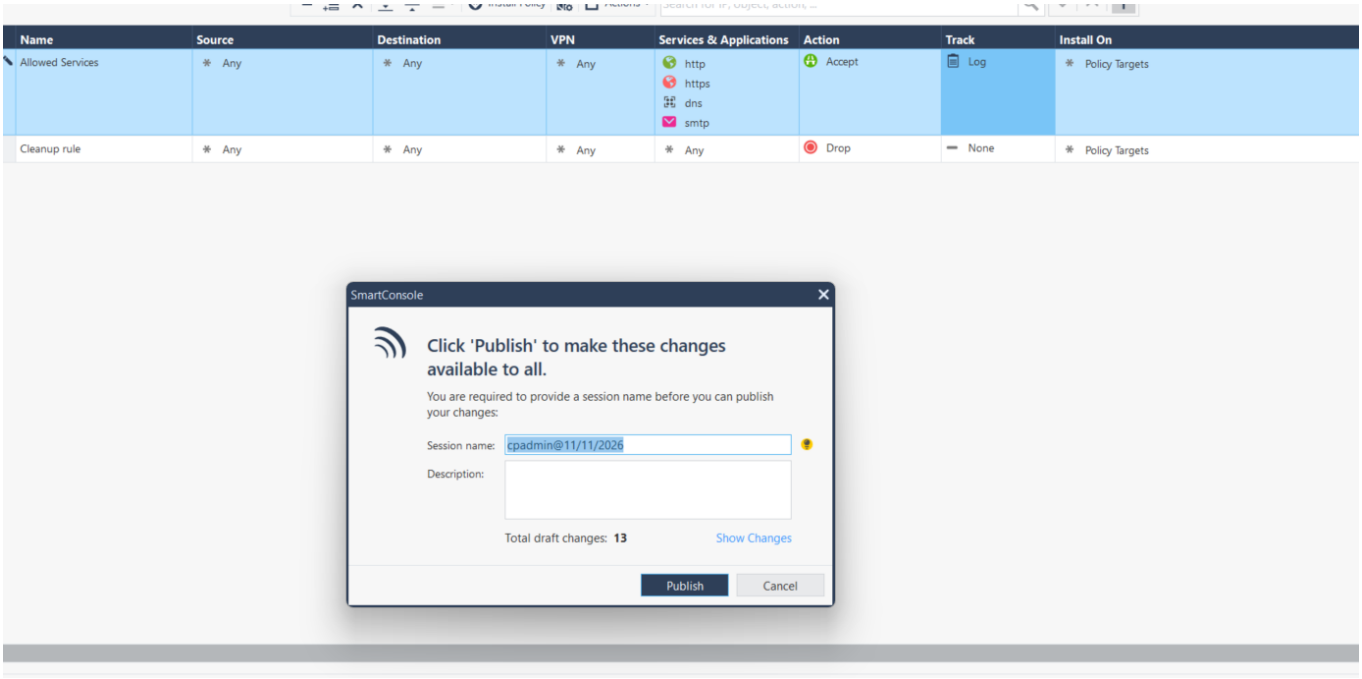
Rule 2: Cleanup Rule

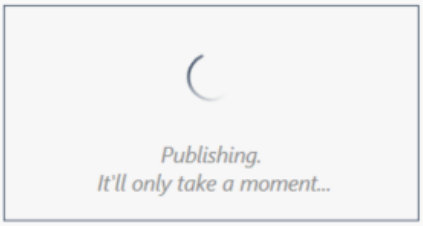


Verifying the Service-Control rulebase



Publishing

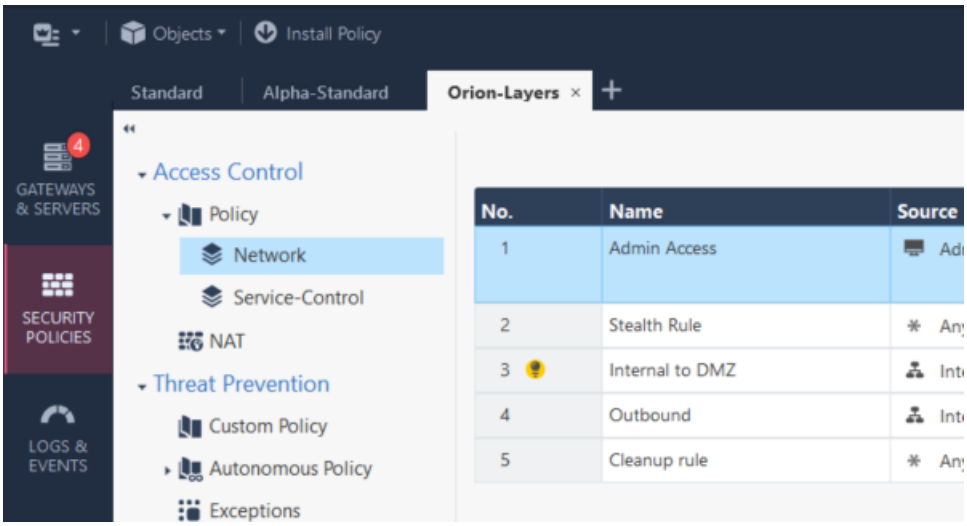

Name
Source
Destination
VPN
Services & Applications
Action
Track
Install On
Allowed Services
Any
Any
Any
http
https
dns
smtp
Accept
Log
Policy Targets
Cleanup rule
Any
Any
Any
Any
Drop
None
Policy Targets
SmartConsole
Click 'Publish' to make these changes available to all.
You are required to provide a session name before you can publish your changes:
Session name:
cpadmin@11/11/2026
Description:
Total draft changes: 13
Show Changes
Publish
Cancel


Publishing.
It'll only take a moment...

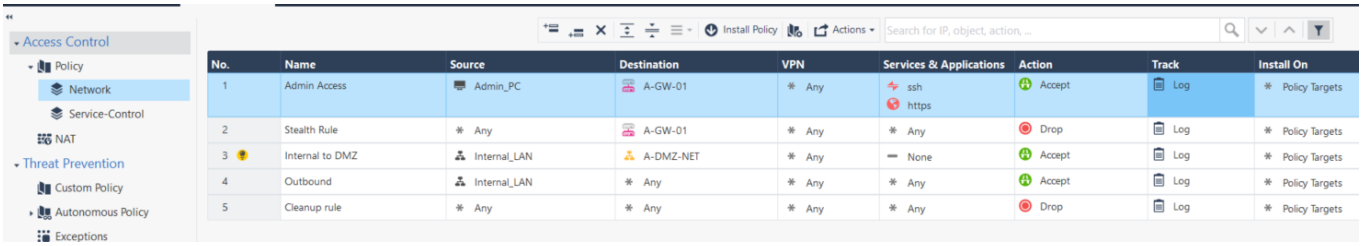
Task 4: Installing, Testing and Fixing

Verifying the policy before installation

1. Confirming the Ordered Layer sequence

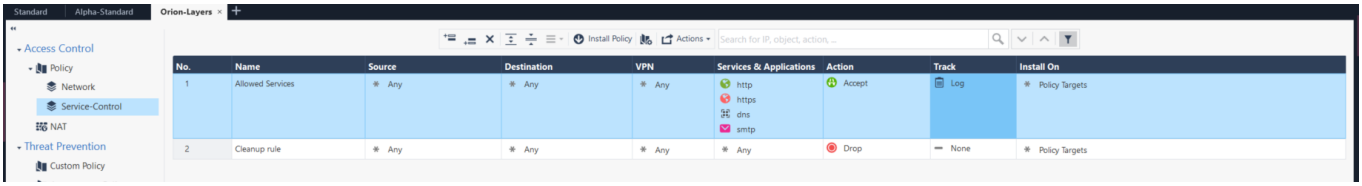


2. Confirming the Network layer



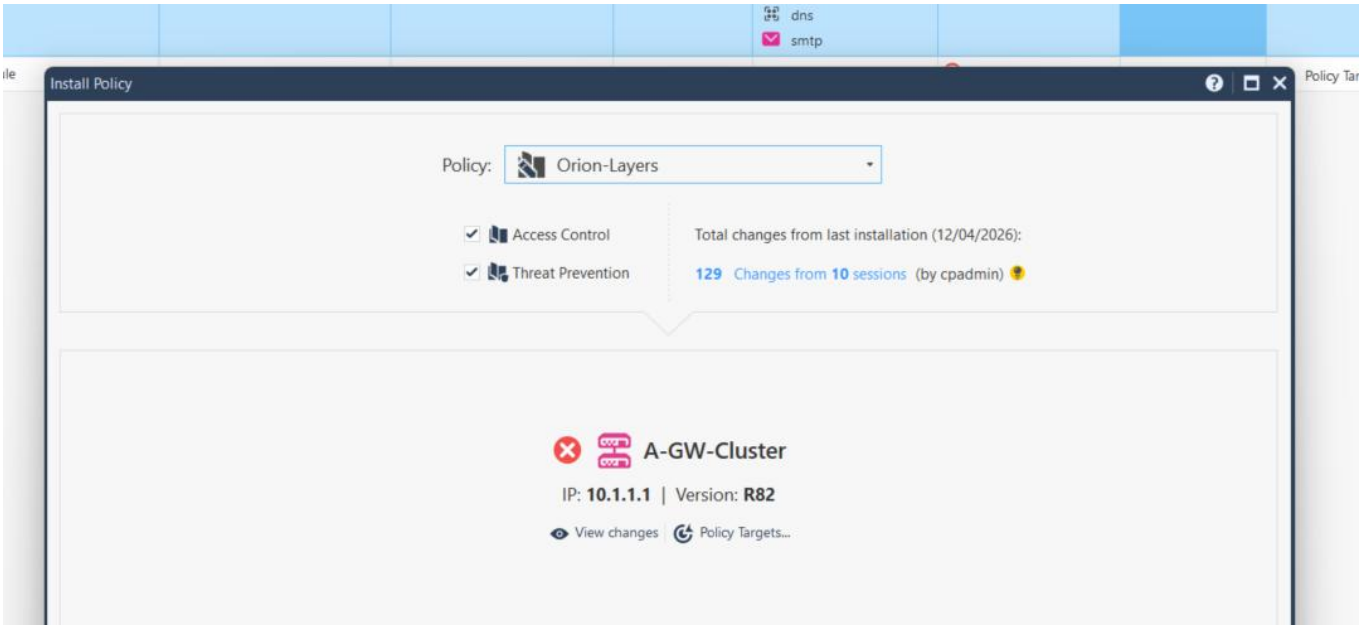
3. Confirming the initial Service-Control layer

Security Policies ⟶ Orion-Layers ⟶ Service-Control

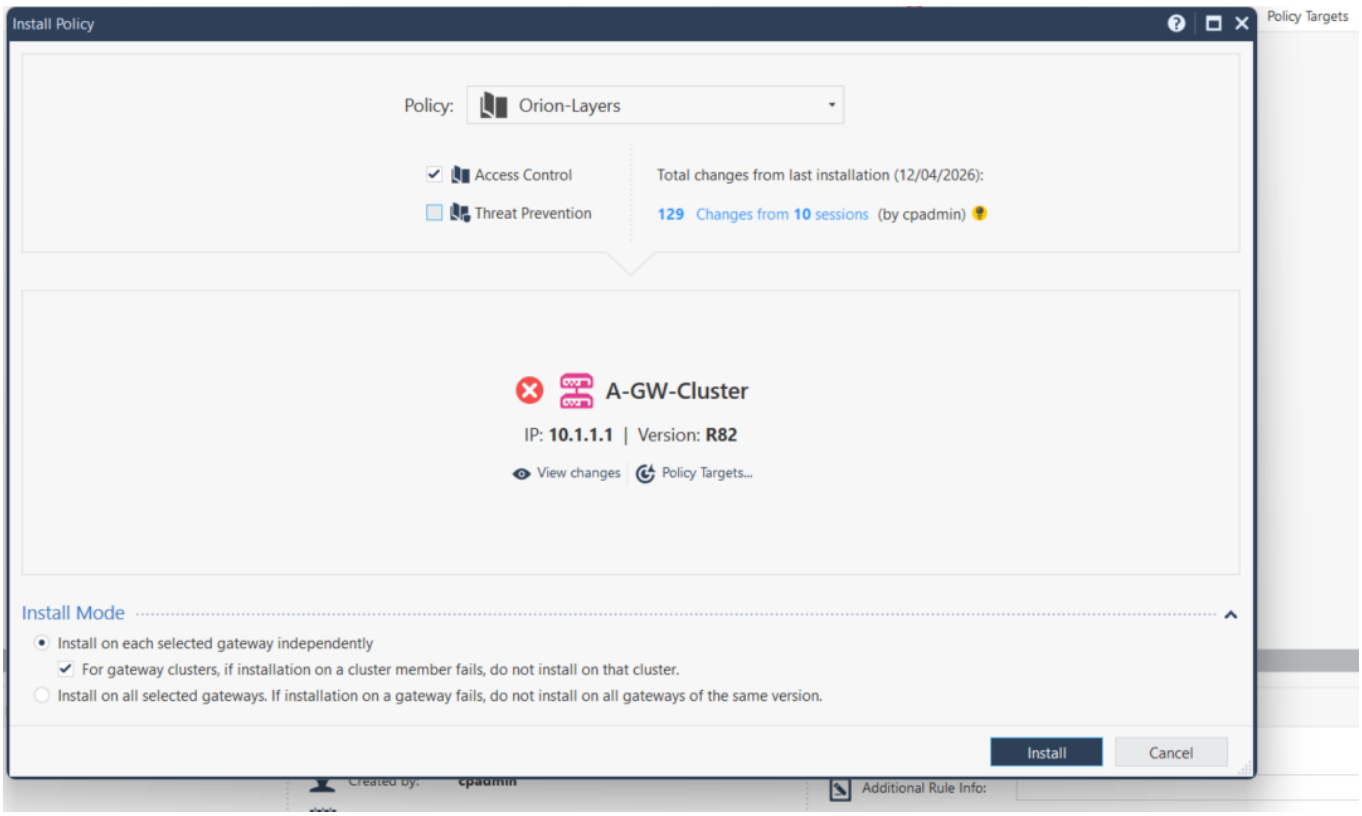


Installing Orion-Layers for the first time

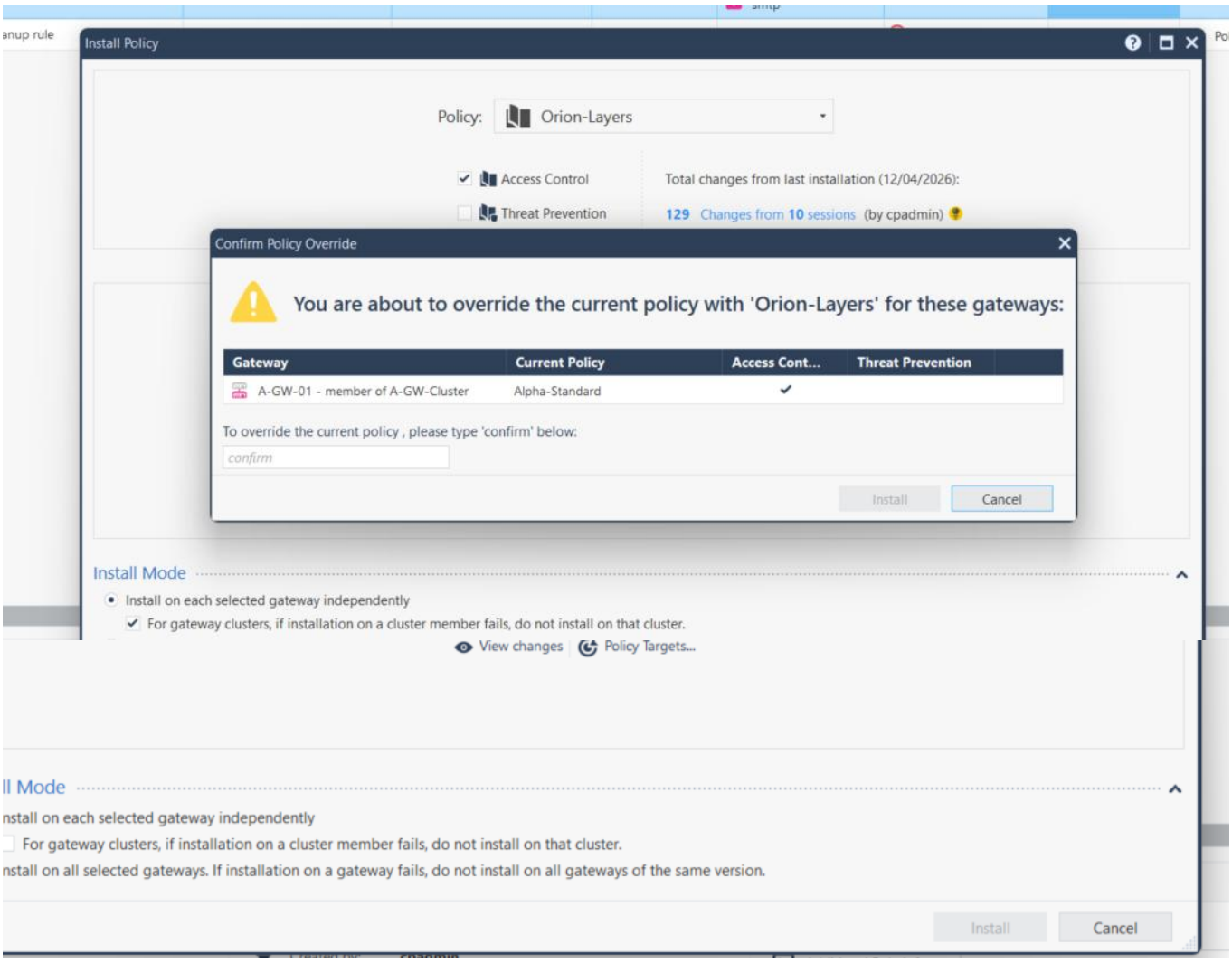
1. Open Install Policy

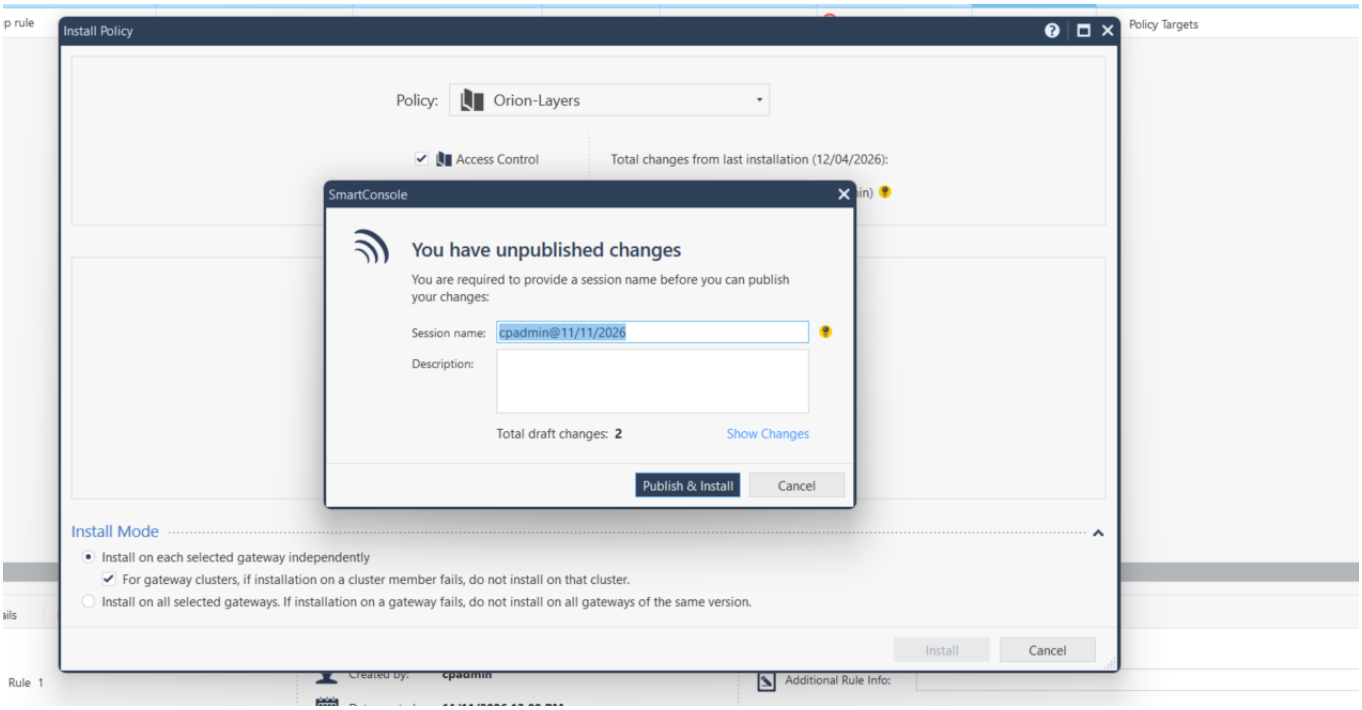


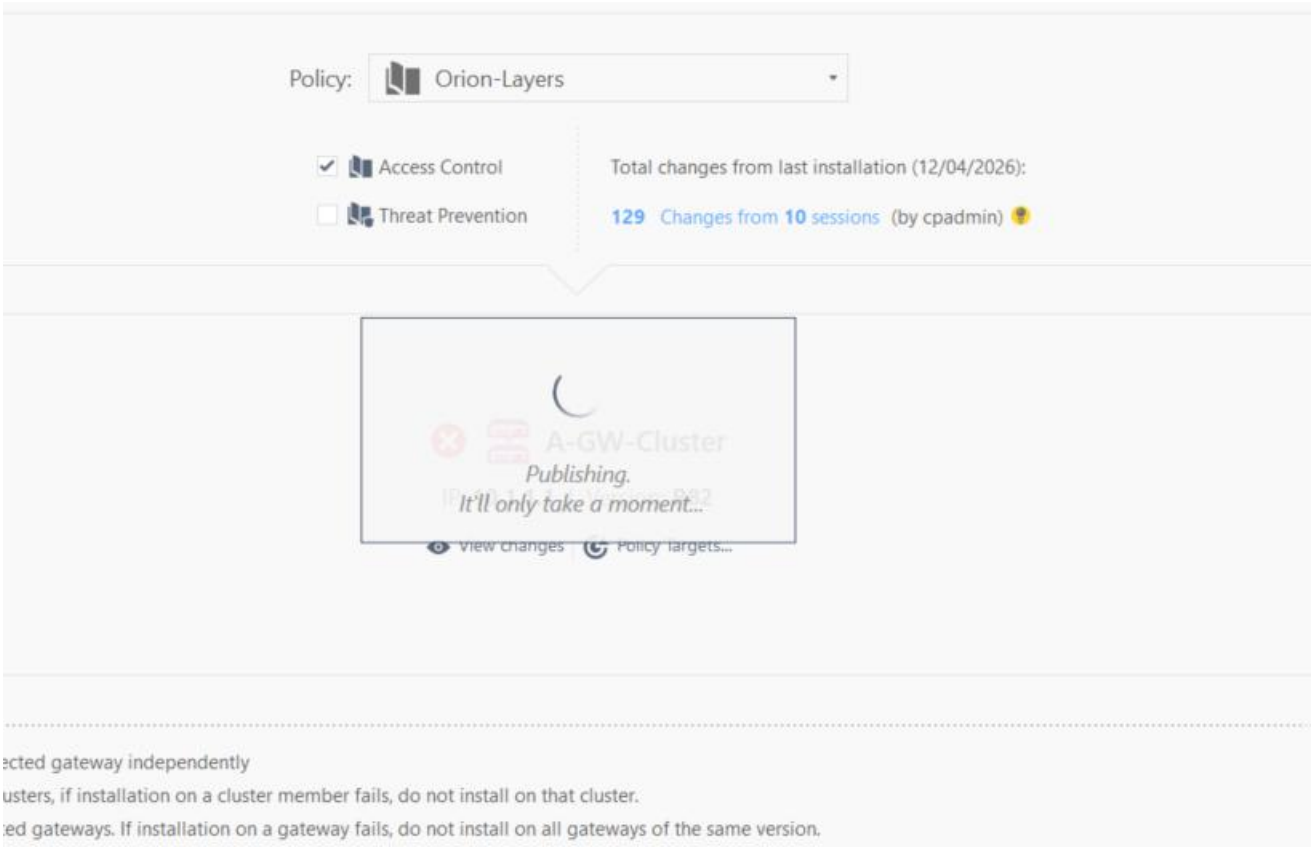
2. Selecting the policy component



Installing


Install Policy
Policy:
Orion-Layers
Access Control
Threat Prevention
Total changes from last installation (12/04/2026):
129 Changes from 10 sessions (by cpadmin)
Confirm Policy Override
You are about to override the current policy with 'Orion-Layers' for these gateways:
Gateway
Current Policy
Access Cont...
Threat Prevention
A-GW-01 - member of A-GW-Cluster
Alpha-Standard
To override the current policy , please type 'confirm' below:
confirm
Install
Cancel
Install Mode
Install on each selected gateway independently
For gateway clusters, if installation on a cluster member fails, do not install on that cluster.
View changes
Policy Targets...
For gateway clusters, if installation on a cluster member fails, do not install on that cluster.
Install on all selected gateways. If installation on a gateway fails, do not install on all gateways of the same version.
Install
Cancel



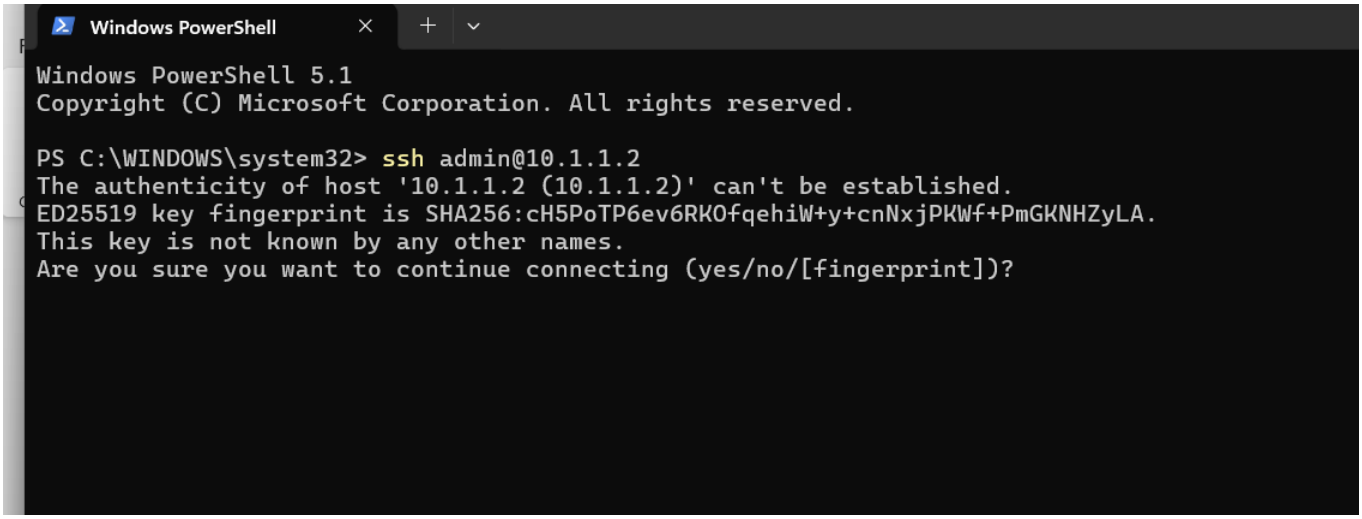


2. Run SSH from Windows

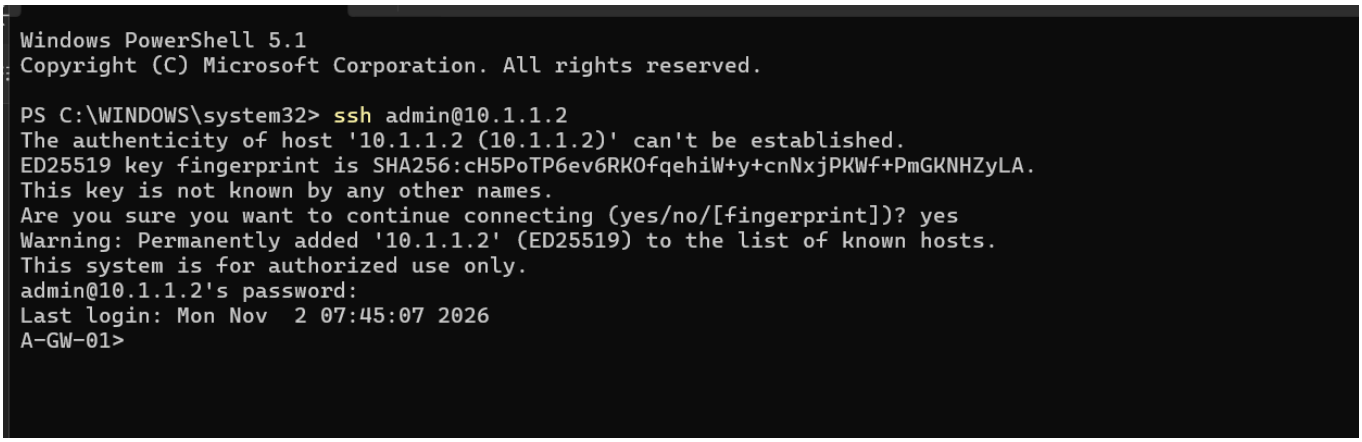
Opening PowerShell on main PC

PowerShell

ssh admin@10.1.1.2



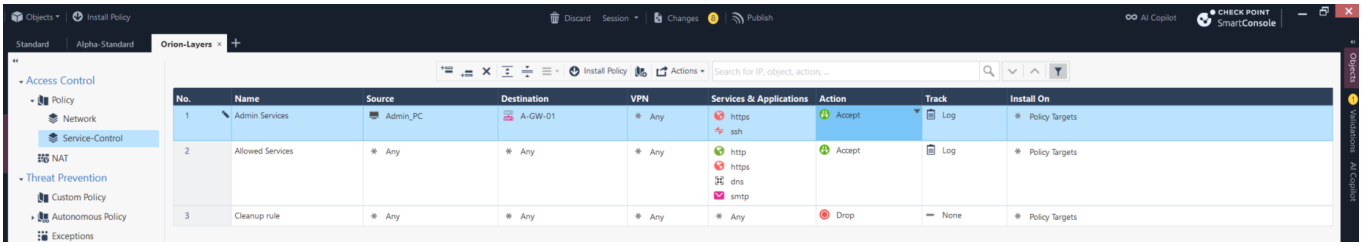
3. Recording the initial result



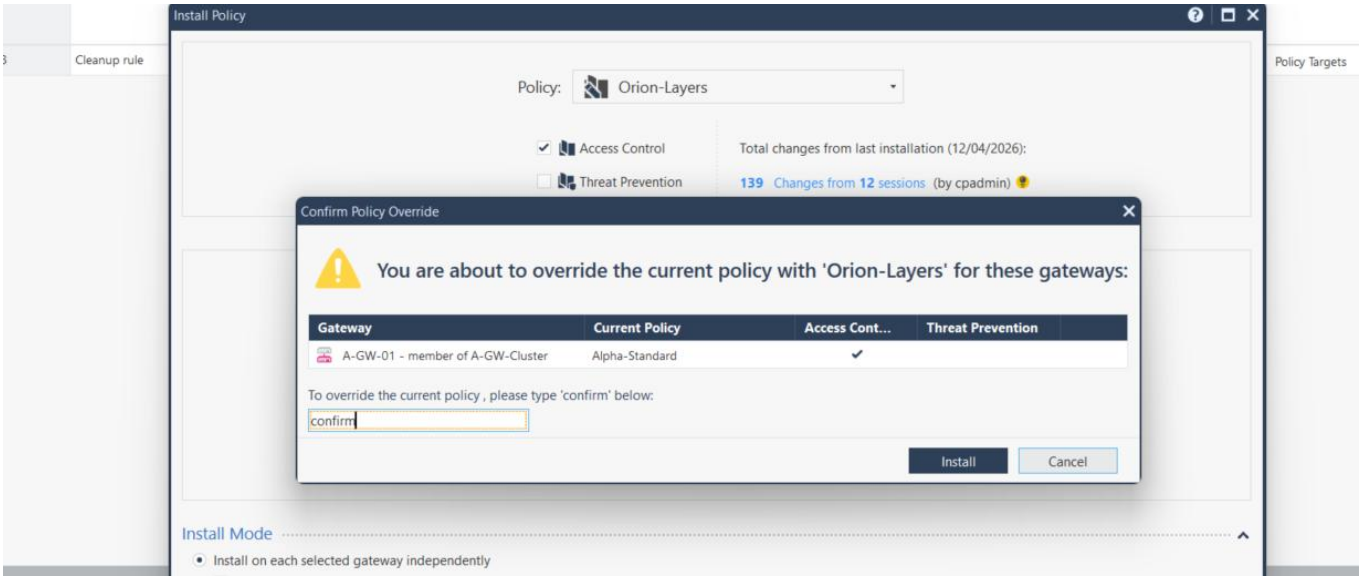
Initial T1 SSH log showing the layer and rule that matched before the Service-Control correction.

The Network layer accepted the SSH connection through the Admin Access rule, but an Accept in one Ordered Layer does not end inspection. The connection continued to Service-Control, where SSH was not included in Allowed Services, so the Service-Control Cleanup Rule dropped the traffic.

Adding the Admin Services correction



Publishing and reinstalling

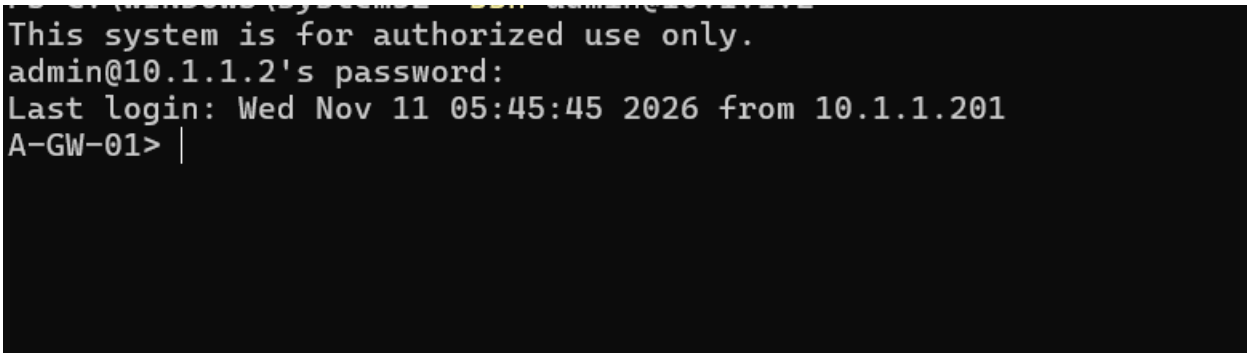


Running T1 again after correction

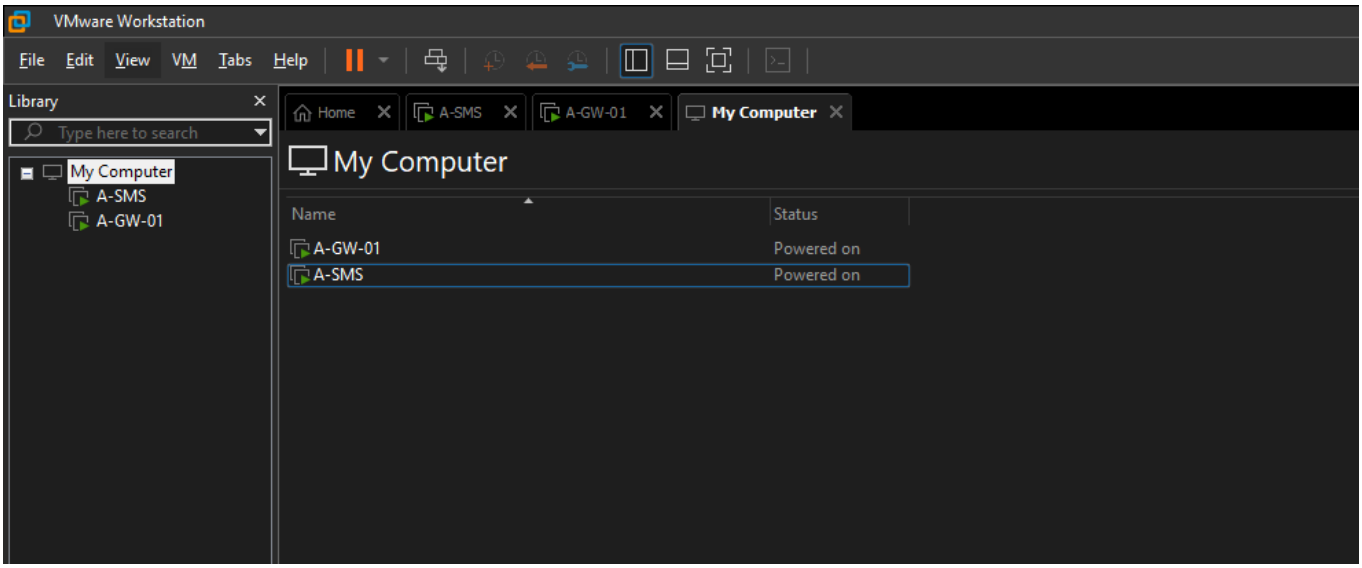
From Admin_PC, run the same SSH command:

PowerShell

ssh admin@10.1.1.2

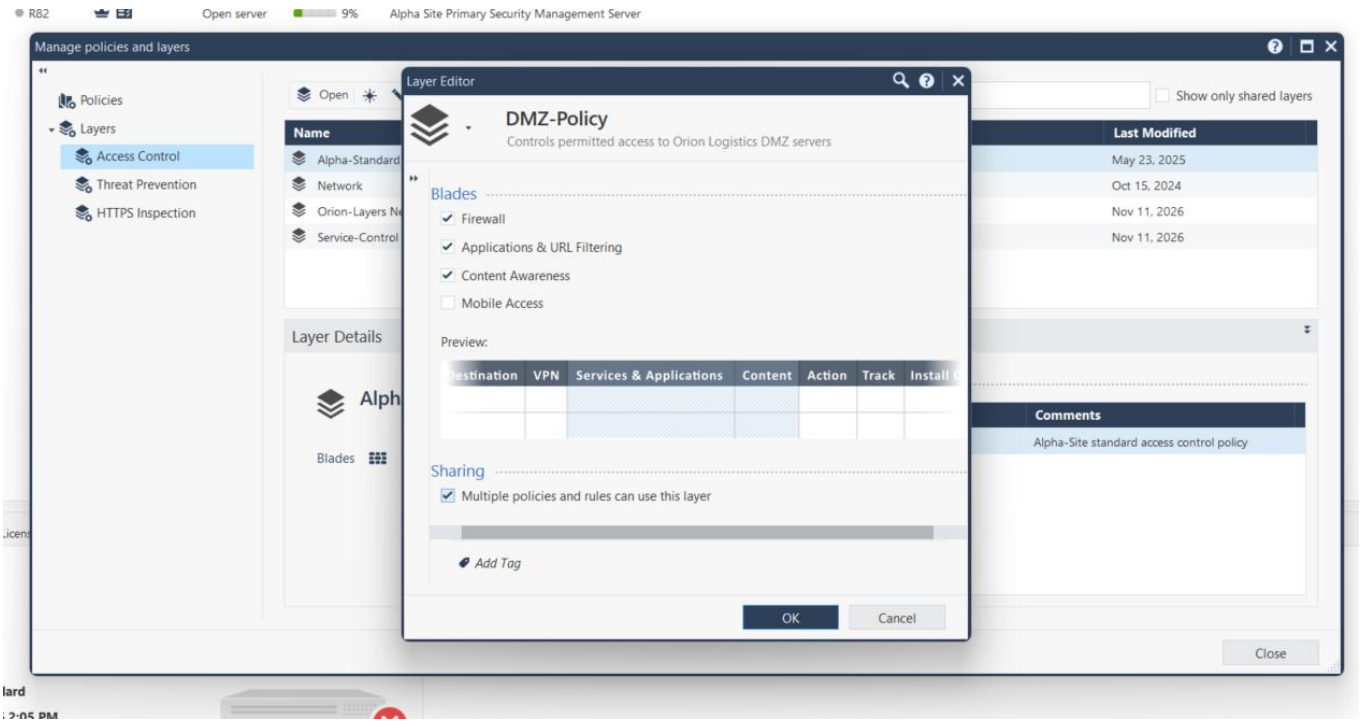


Due Processing power , the Internal Vm were not added to perform all the test

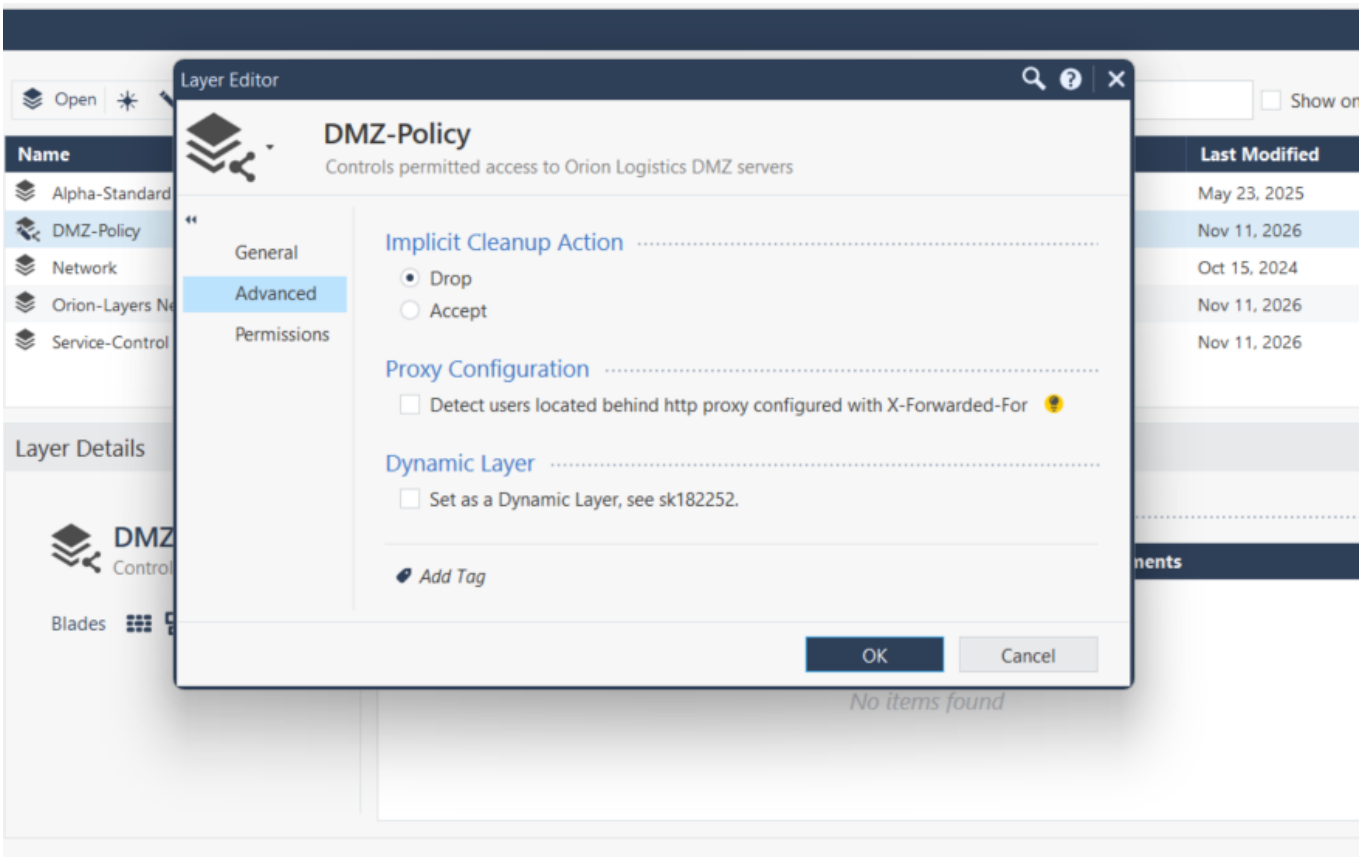


Task 5: DMZ-Policy Inline Layer

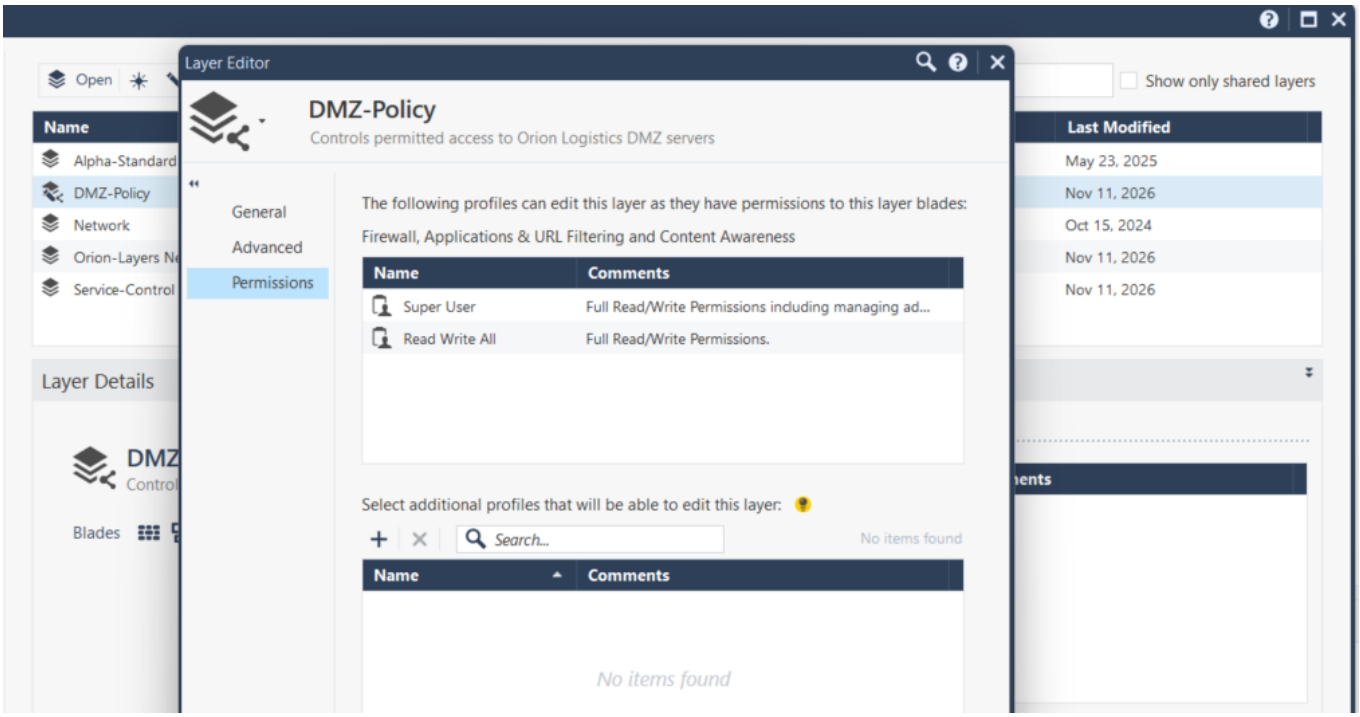
Configuring the General page



Configuring the Advanced page

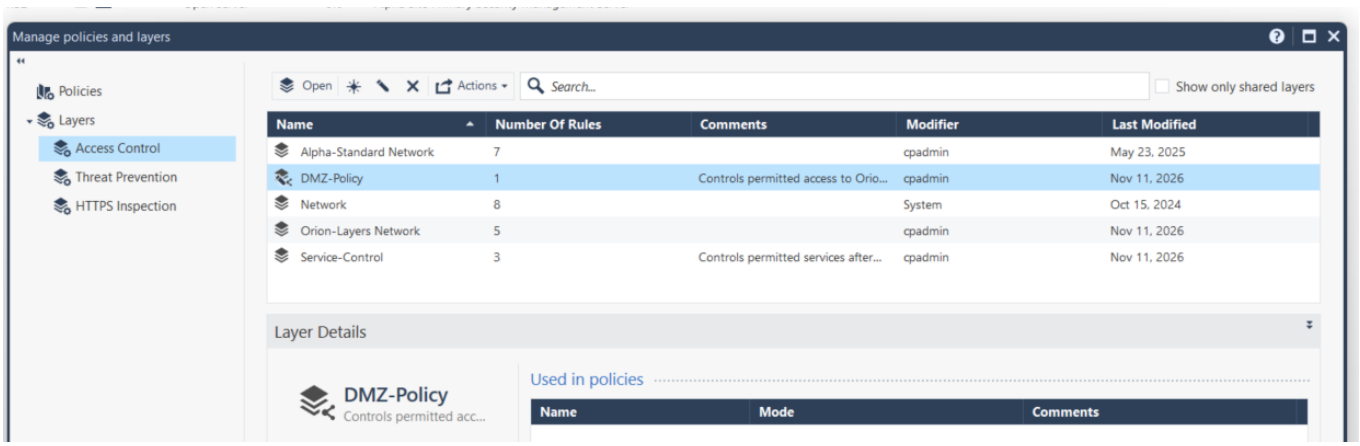


Configure Permissions

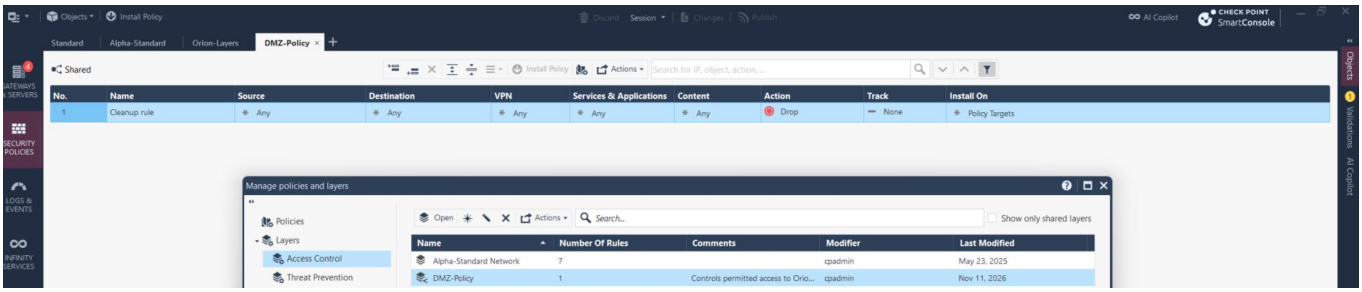


192

Save the layer

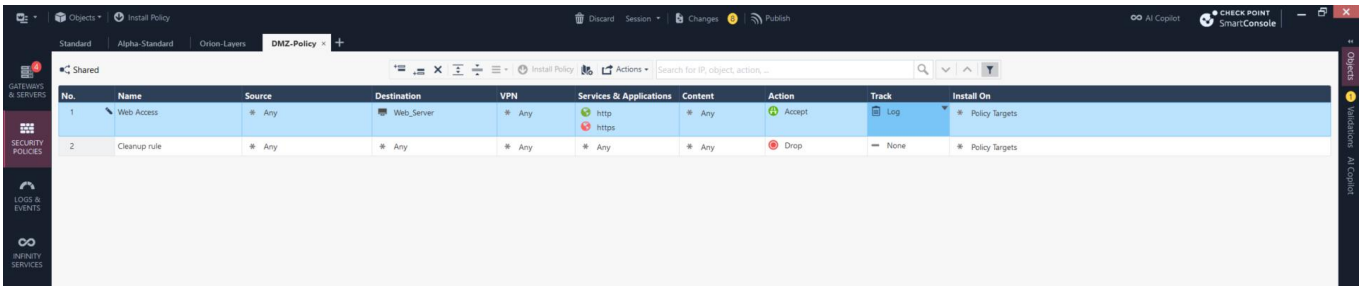


Open the DMZ-Policy rulebase



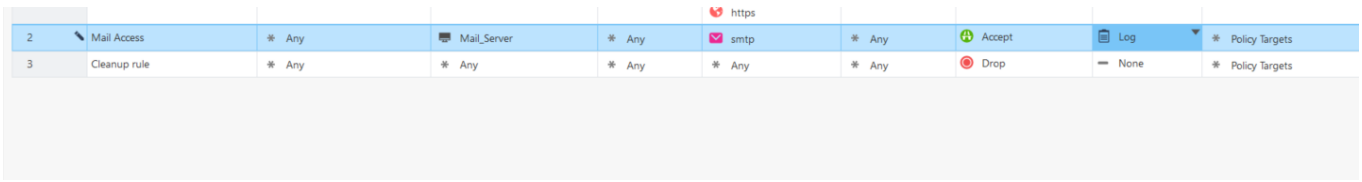
Building the rulebase

Rule 1: Web Access

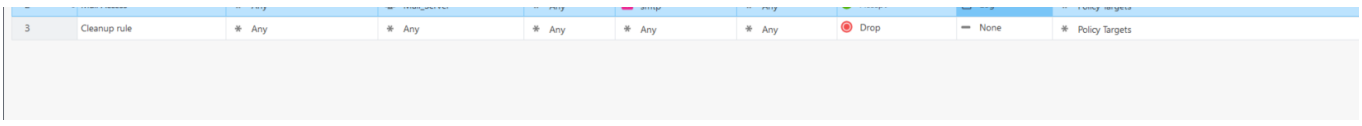


Rule 2: Mail Access

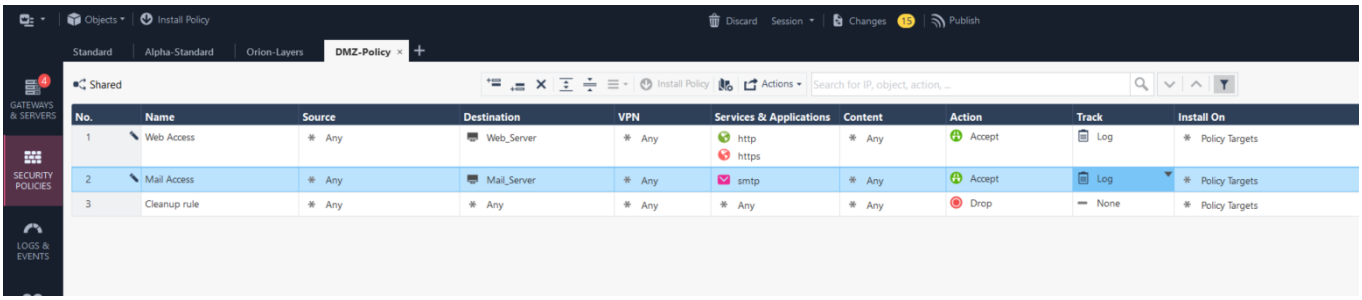
Step 17: Configure Mail Access



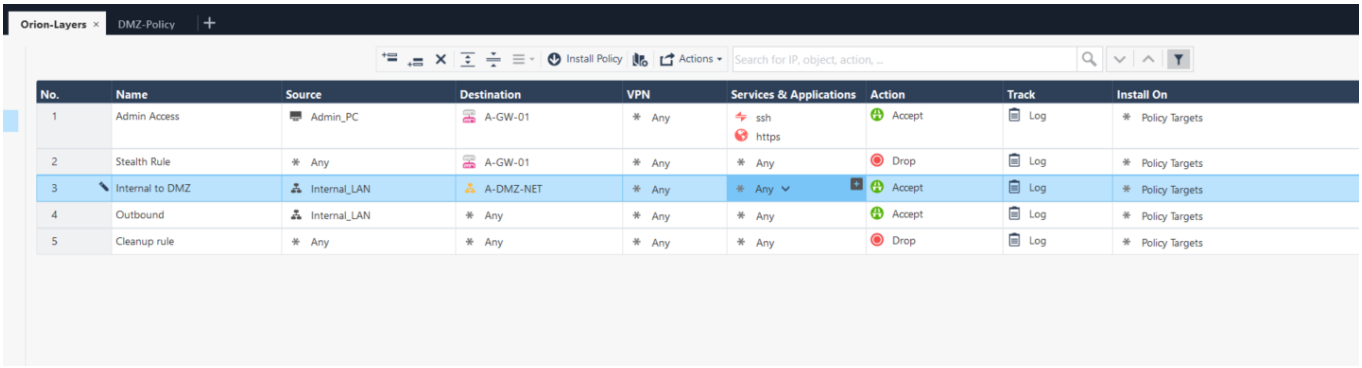
Rule 3: Cleanup Rule



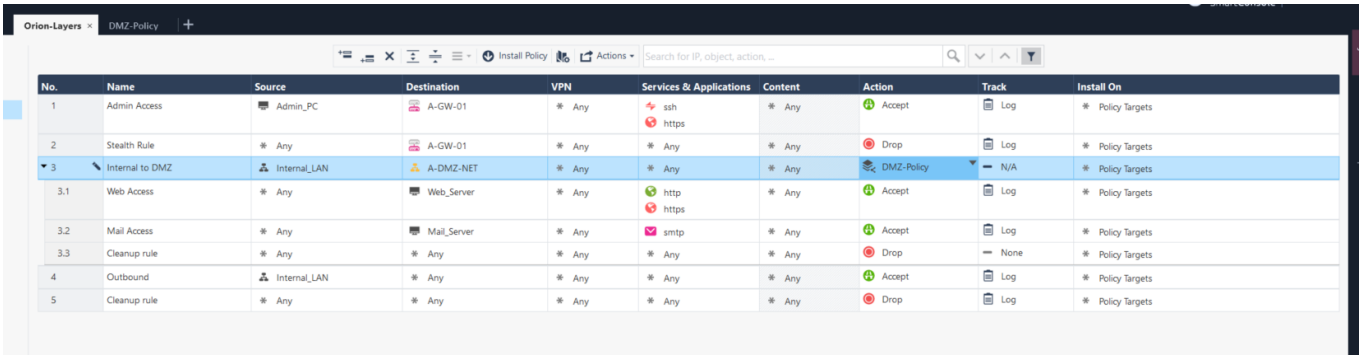
Checking the DMZ-Policy rulebase



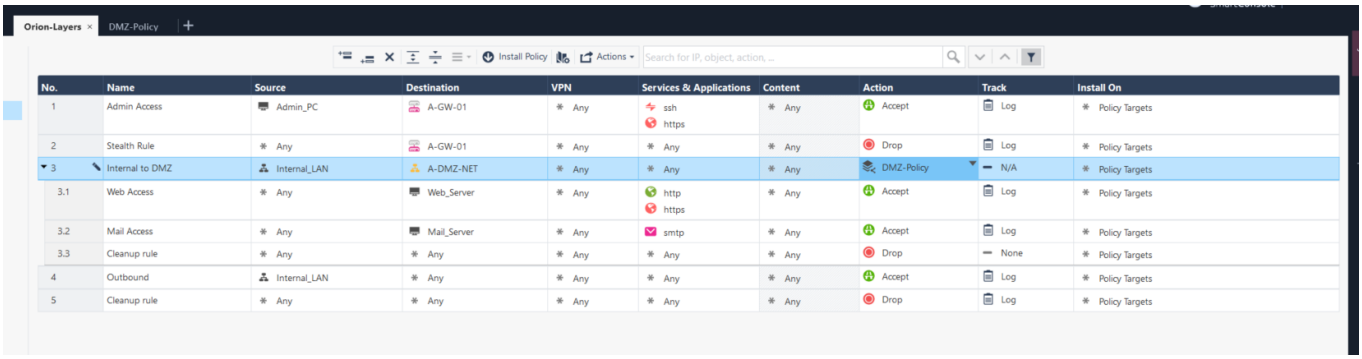
Changing Network Rule 3 to use the Inline Layer



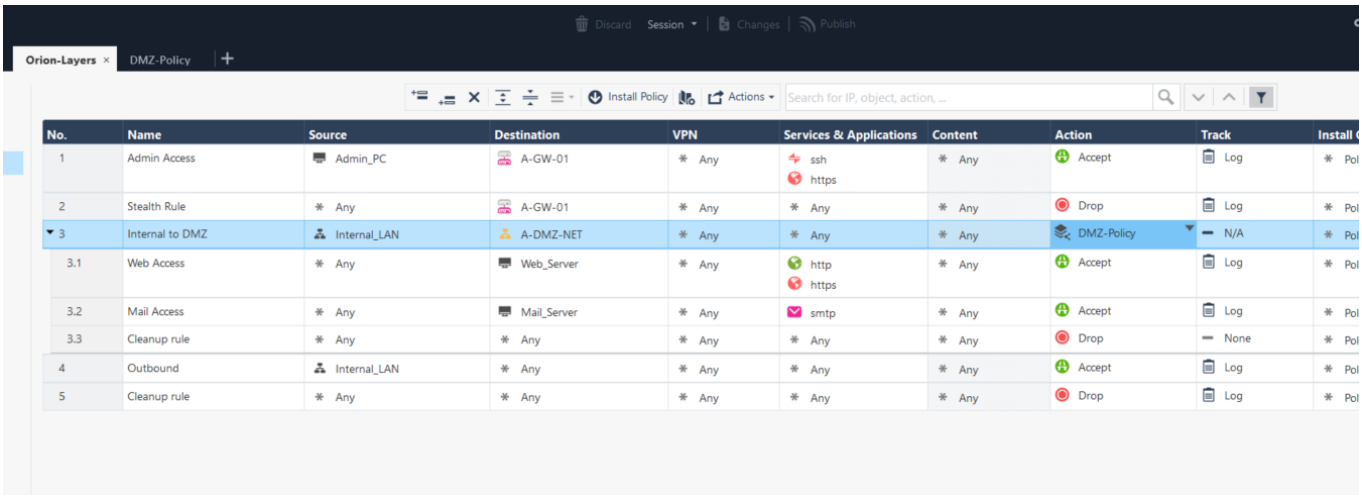
Changing the Action



Verifying Rule 3



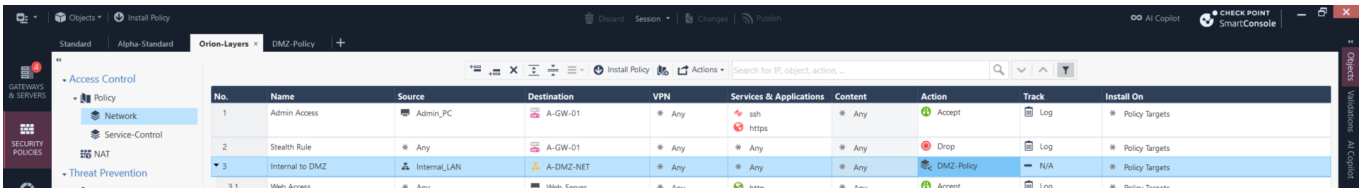
Reviewing and publishing



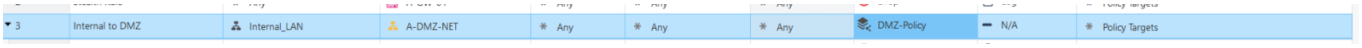
Task 6: Installing and Verifying the DMZ-Policy Inline Layer

Final checks before installation

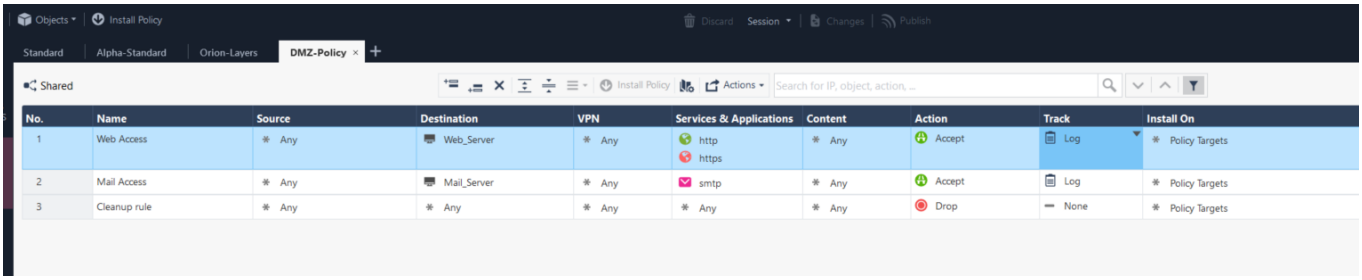
1. Verifying the Ordered Layers



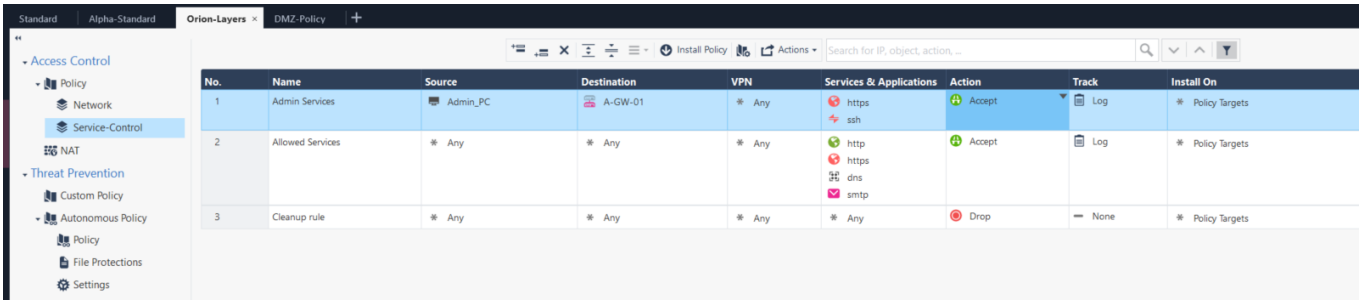
2. Verifying the Network layer



3. Verifying DMZ-Policy

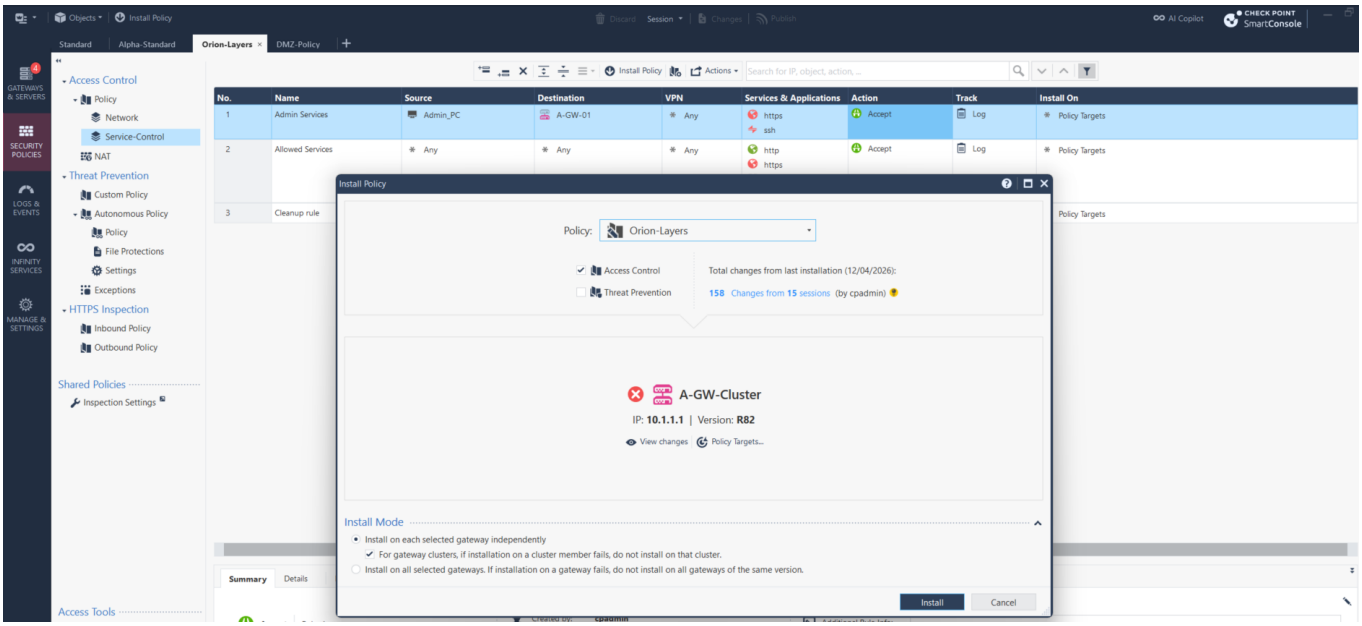


4. Verifying Service-Control

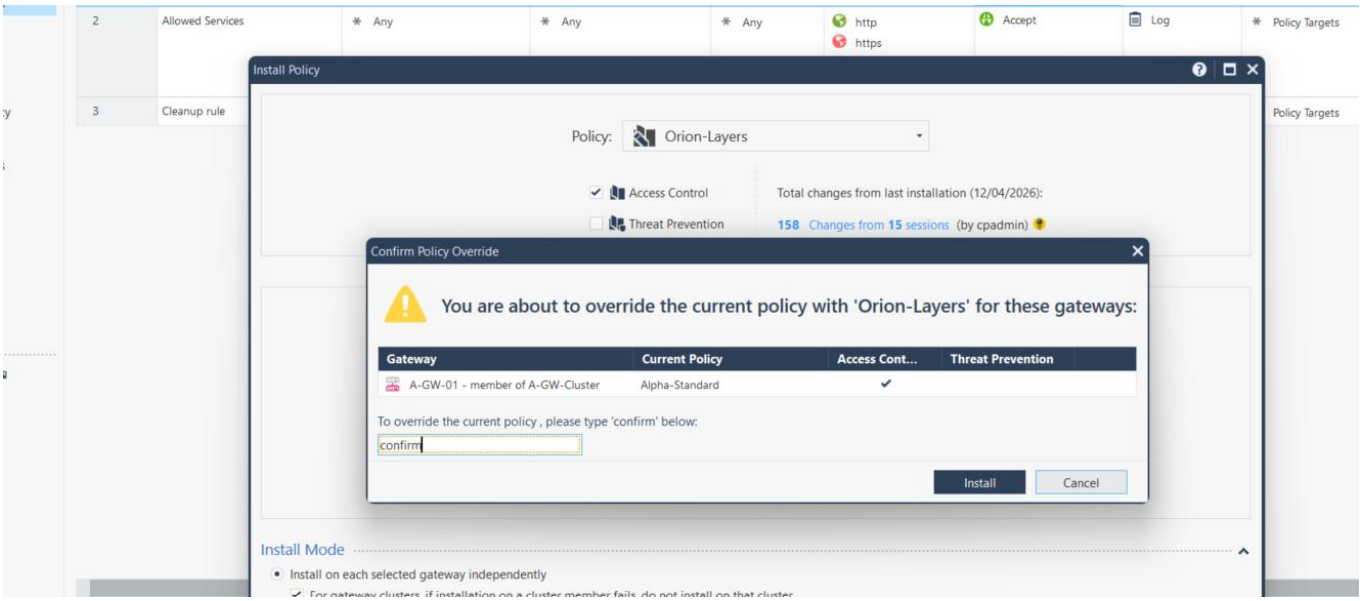


Installing Orion-Layers

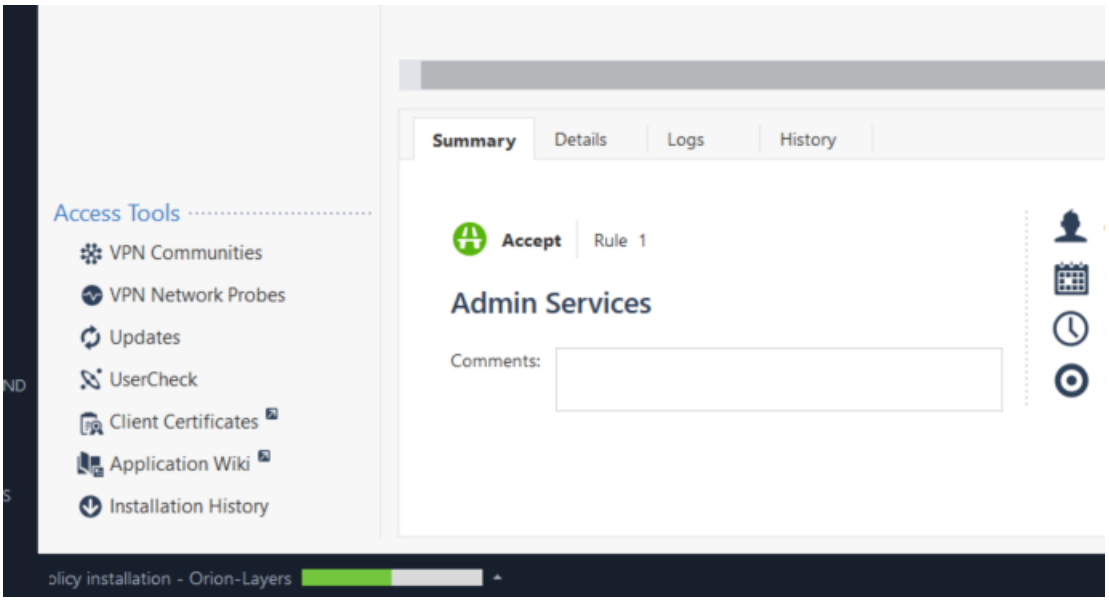
1. Start policy installation

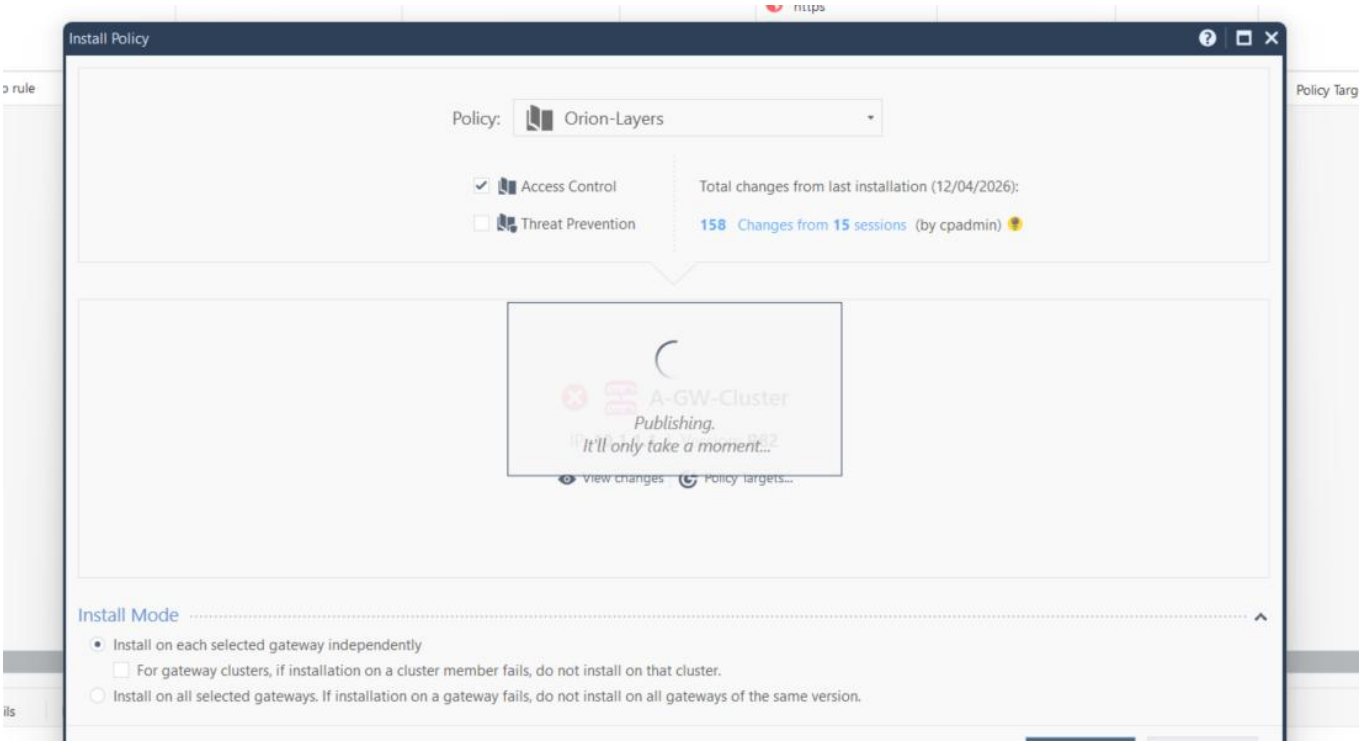


2. Selecting the right gateway

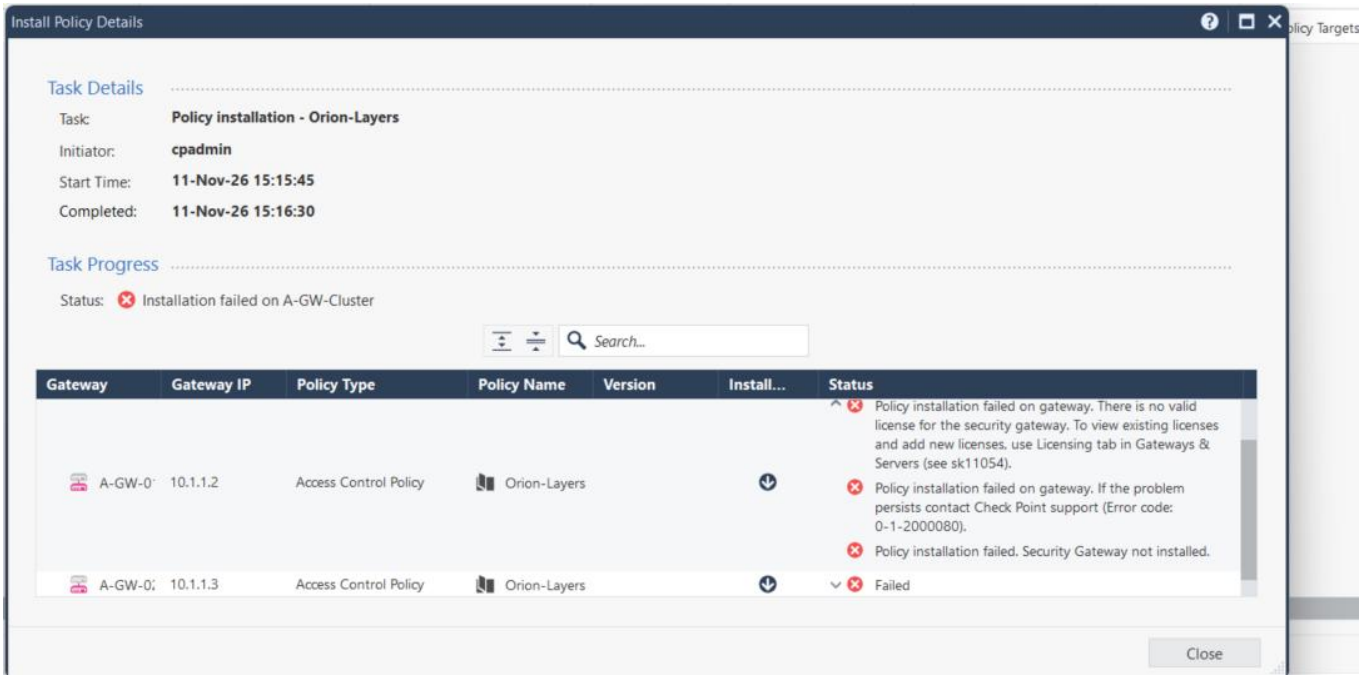


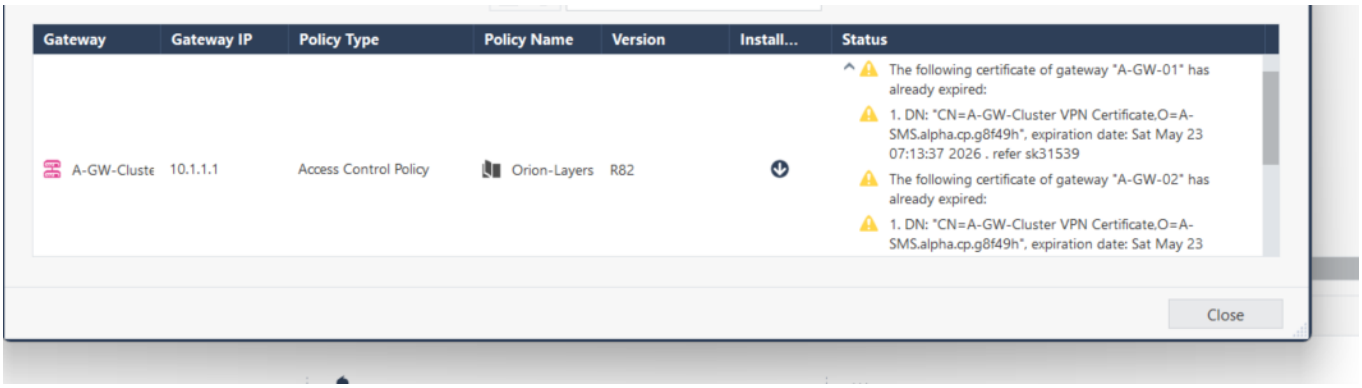
Installation evidence





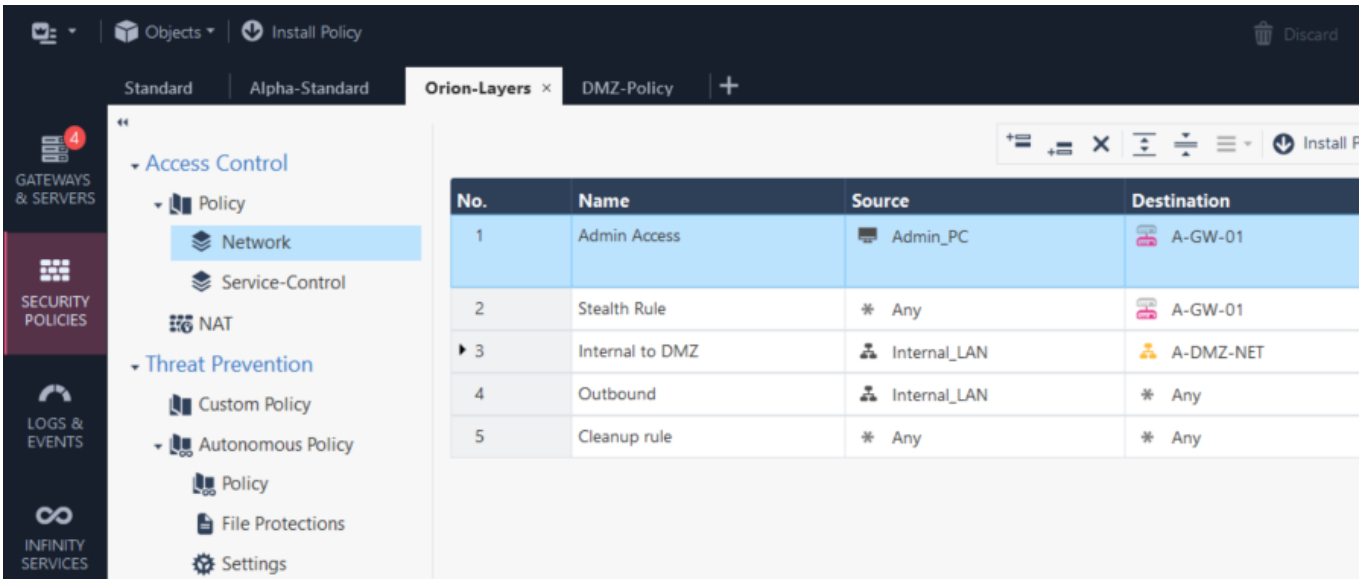
Due to license



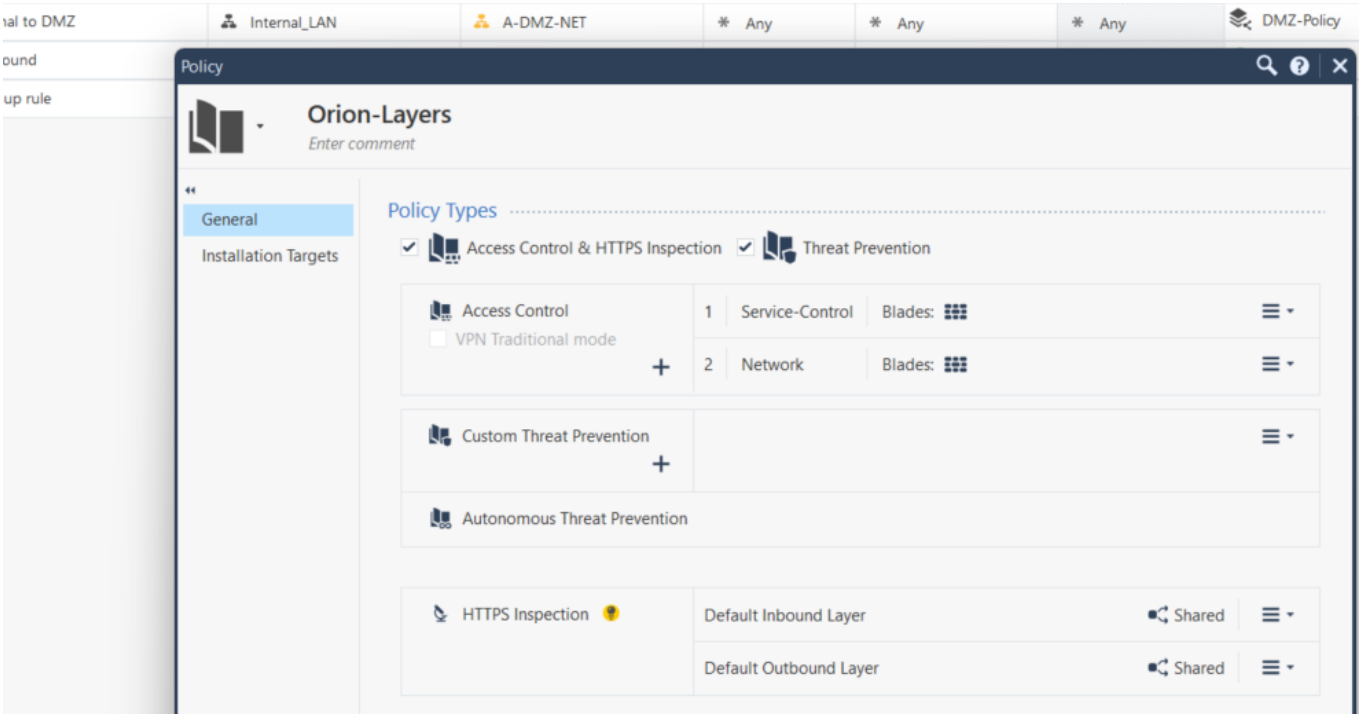


Task 7: Changing Ordered-Layer Order and Verify Sharing

Recording the original layer order

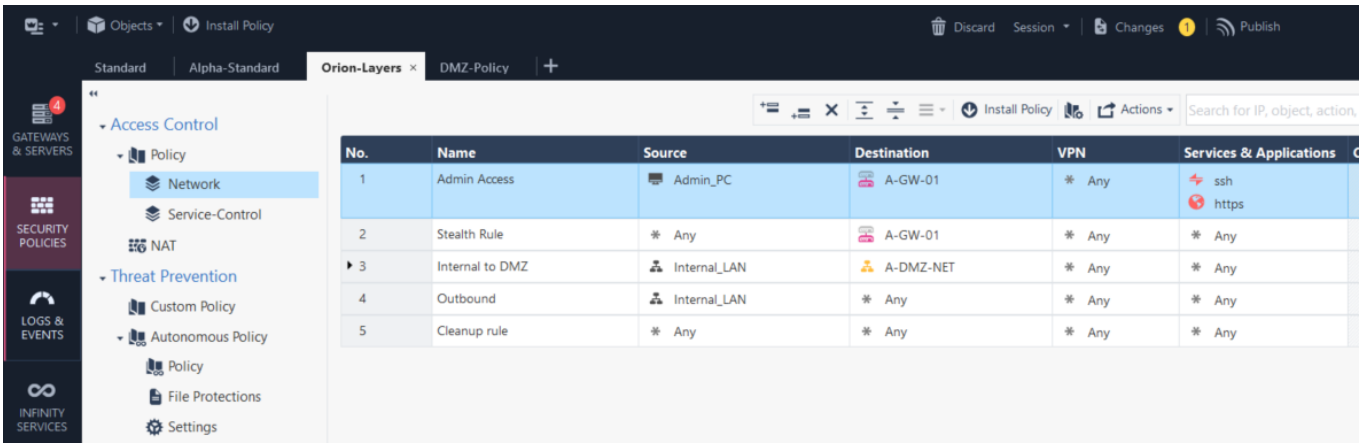


Moving Service-Control above Network



Installing the temporary policy order

Restoring the original order.



The Sharing setting makes DMZ-Policy available for use in more than one policy package. Another package could select the existing DMZ-Policy Inline Layer instead of creating a separate copy of the Web Access, Mail Access and Cleanup rules. This promotes consistent DMZ controls and reduces duplicated administration.

4. Part B: Theory Answers

Section B1: Multiple Choice

Question	Answer	Explanation
Q1	B	A policy layer is a set of rules that divides a policy into smaller, more manageable rulebases.
Q2	C	The default Access Control policy is Standard and its default Ordered Layer is Network.
Q3	A	An Accept in one Ordered Layer continues to the next Ordered Layer; all applicable layers must permit the traffic.
Q4	D	An Inline Layer is attached to a specific rule and is evaluated as a sub-policy when that parent rule matches.
Q5	B	An explicit Cleanup rule with the same action as the Implicit Cleanup rule makes the final decision visible and loggable.

Section B2: Short Answers

Q6. Definition and benefits of policy layers

A policy layer is a separate rulebase within an Access Control policy that handles a defined part of traffic inspection. Layers are useful in large networks because they divide a long policy into smaller logical controls. Two benefits are easier review and maintenance, and the ability to assign responsibility for specific layers through permission profiles.

Q7. Ordered Layers and Inline Layers

Ordered Layers are evaluated sequentially in the order listed in the policy package. An Accept in an earlier Ordered Layer allows processing to continue to the next layer, while a Drop stops the connection; a suitable use case is separating network access from allowed-service control. An Inline Layer is attached to the action of a specific parent rule and is evaluated only when that parent rule matches; a suitable use case is applying a dedicated DMZ sub-policy only to traffic between the Internal and DMZ networks.

Q8. Enforcement order

The gateway evaluates Ordered Layers from top to bottom. A Drop action ends evaluation and blocks the connection, while an Accept allows the connection to continue to the next Ordered Layer. If no explicit rule matches, the layer's Implicit Cleanup Action applies, which is why a later layer with an implicit Drop can block traffic accepted by an earlier layer.

Q9. Layer setting groups

The three layer-setting groups are General, Advanced and Permissions. General includes settings such as the layer name, enabled blades and Sharing; Advanced includes the Implicit Cleanup Action, Proxy Configuration and Dynamic Layer; Permissions identifies the permission profiles that can edit the layer, such as Super User and Read-Write All.

Q10. Sharing

Sharing determines whether a layer can be reused in more than one policy package. A shared layer promotes consistency and reduces duplication, but it should contain controls that are genuinely common because a change to the shared layer can affect every package in which it is used.

Section B3: Scenario Questions

Q11. Accepted in the first layer but unmatched in the second

The connection is dropped because the Accept in the first Ordered Layer continues inspection into the second layer, where no explicit rule matches and the Implicit Cleanup Action is Drop. I would confirm the decision in Logs & Events by checking the matching layer, rule or implicit cleanup information, source, destination, service and action. If the connection should be allowed, I would add a specific Accept rule in the second layer for the required traffic, publish and install the policy, and retest. An explicit Cleanup rule is recommended because it makes the final action visible in the rulebase, can be configured for logging and is easier to audit than relying only on an implicit decision.

Q12. Layer design for a large policy

I would separate common network-level decisions into a first Ordered Layer, place service and application restrictions in a second Ordered Layer, and attach Inline Layers to parent rules for specialised areas such as DMZ, remote access or sensitive applications. This design shortens each rulebase, lets reviewers focus on one control area, and allows permission profiles to assign administrators only the layers relevant to their responsibilities. Performance and troubleshooting also improve because broad network decisions can be made early, while specialised Inline Layers are evaluated only when their parent rules match; however, rule order, cleanup actions and shared-layer dependencies must be documented to avoid unintended drops.

Gateway licensing limitation

SmartConsole displayed an expired licence warning for the supplied lab gateways. Although the instructor confirmed that a separate licence was not required for the laboratory exercise, the completed changes could be published to the management database but the Orion-Layers policy could not be successfully installed on the gateway because of the licence status. Consequently, tests requiring the newly installed policy and the related log evidence could not be completed. The configuration was retained in SmartConsole, and no attempt was made to add, copy or modify a licence.

Issue or limitation	Effect on the assignment	Action taken
The supplied gateways displayed expired licence warnings, although the instructor confirmed that no additional licence was required.	The Orion-Layers policy could not be successfully installed, so the remaining traffic tests and matching log evidence could not be completed.	The error was recorded honestly. No licence was copied, added or modified, and no test results were fabricated.

8. References

1. CCSA105 Practical Assignment: Policy Layers.
2. CCSA105 Module 5 slides.
3. Companion Lab 5A: Configure Ordered Layers.
4. Companion Lab 5B: Configure Inline Layers.
5. Check Point Quantum Security Management Administration Guide.